Topology I

Lecture notes, Chapters 1–8 (General Topology)

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Chapter 1. Set Theory and Logic

This chapter collects the set theory used throughout. Numbers in parentheses after a result refer to the corresponding result in Munkres.

1.1 Fundamental concepts

Motivation. Topology needs very little machinery beyond sets and functions, but it uses that machinery heavily: unions and intersections of arbitrary families, images and preimages, and infinite constructions. This chapter fixes the language and proves the set-theoretic facts (countability, choice, well-ordering) that later arguments quietly rely on.

We use sets, membership, inclusion $A \subset B$, union, intersection, difference $A - B$, the empty set, and the cartesian product $A \times B$ of ordered pairs. For a collection $\mathcal{A}$ of sets, $\bigcup_{A\in\mathcal{A}} A$ and $\bigcap_{A\in\mathcal{A}} A$ are defined as usual (the intersection of the empty collection is left undefined). De Morgan's laws hold:

$$A - \bigcup_\alpha B_\alpha = \bigcap_\alpha (A - B_\alpha), \qquad A - \bigcap_\alpha B_\alpha = \bigcup_\alpha (A - B_\alpha),$$

and $A \cap \bigcup B_\alpha = \bigcup (A \cap B_\alpha)$, $A \cup \bigcap B_\alpha = \bigcap (A \cup B_\alpha)$. In logic, "$P \Rightarrow Q$" is equivalent to its contrapositive "not $Q \Rightarrow$ not P" but not to its converse; the negation of "for every x, $P(x)$" is "for some x, not $P(x)$".

1.2 Functions

A **function** $f : A \to B$ is a rule (formally a subset of $A\times B$ in which each a appears as first coordinate of exactly one pair). For $A_0 \subset A$, $f(A_0)$ is the image; for $B_0 \subset B$, $f^{-1}(B_0) = \{a : f(a) \in B_0\}$. f is **injective** if $f(a) = f(a')$ implies $a = a'$, **surjective** if $f(A) = B$, **bijective** if both; a bijection has an inverse $f^{-1} : B \to A$. Composites of injections (surjections) are injections (surjections). Preimages preserve unions, intersections and differences; images preserve unions and inclusions only, and $A_0 \subset f^{-1}(f(A_0))$, $f(f^{-1}(B_0)) \subset B_0$, with equality if f is injective (resp. surjective).

Lemma 1.1 (2.1)

Let $f : A \to B$. If there are $g, h : B \to A$ with $g(f(a)) = a$ for all $a \in A$ and $f(h(b)) = b$ for all $b \in B$, then f is bijective and $g = h = f^{-1}$.

Proof. $f(a) = f(a')$ gives $a = g(f(a)) = g(f(a')) = a'$, so f is injective; $b = f(h(b))$ shows it is surjective. Then $g = g \circ (f \circ h) = (g \circ f) \circ h = h$, and both equal f^{-1}. ∎

1.3 Relations

Motivation. Equivalence relations will produce quotient spaces, and order relations will produce the order topology on $\mathbb{R}$ and on well-ordered sets. Both are introduced here in purely set-theoretic form.

A **relation** on A is a subset $C \subset A \times A$, written xCy. An **equivalence relation** $\sim$ is reflexive, symmetric and transitive; the **equivalence class** of x is $E = \{y : y \sim x\}$.

Lemma 1.2 (3.1)

Two equivalence classes E, E' are either disjoint or equal.

Proof. If $y \in E \cap E'$ with E the class of x and E' the class of x', then $y \sim x$ and $y \sim x'$, so $x \sim x'$. Any $w \sim x$ then satisfies $w \sim x'$; so $E \subset E'$ and symmetrically. ■

Hence the classes form a **partition** of A (disjoint nonempty sets with union A); conversely every partition comes from the relation "lying in the same piece".

An **order relation** (simple or linear order) $<$ satisfies: comparability ($x \neq y \Rightarrow x < y$ or $y < x$), nonreflexivity (no $x < x$) and transitivity. In an ordered set we have open intervals (a, b), immediate successors and predecessors, and the **dictionary order** on $A \times B$: $a_1 \times b_1 < a_2 \times b_2$ iff $a_1 < a_2$, or $a_1 = a_2$ and $b_1 < b_2$. Ordered sets A, B have the same **order type** if there is an order-preserving bijection. For $A_0 \subset A$, an **upper bound** b satisfies $x \leq b$ for all $x \in A_0$; the least one is the **supremum** $\sup A_0$, and dually $\inf A_0$. A has the **least upper bound property** if every nonempty subset bounded above has a supremum; this is equivalent to the greatest lower bound property.

1.4 The integers and the real numbers

Motivation. All of analysis rests on one property of $\mathbb{R}$: every nonempty bounded-above set has a least upper bound. We record the axioms of $\mathbb{R}$ and the induction principle for $\mathbb{Z}_+$, so that later proofs (connectedness of intervals, compactness of $[a, b]$) can point to exactly what they use.

We assume $\mathbb{R}$ is an ordered field with the least upper bound property; $\mathbb{Z}_+$ is the smallest inductive subset (containing 1 and closed under $x \mapsto x + 1$), which gives the principle of induction.

Theorem 1.3 (4.1, well-ordering property)

Every nonempty subset of $\mathbb{Z}_+$ has a smallest element.

Proof. First, for each n, every nonempty subset of $\{1, \ldots, n\}$ has a smallest element: the set A of n for which this holds contains 1, and if $n \in A$ and $C \subset \{1, \ldots, n+1\}$, then either $C = \{n+1\}$ or $C \cap \{1, \ldots, n\}$ has a least element, which is the least element of C. For a nonempty $D \subset \mathbb{Z}_+$ choose $n \in D$; the least element of $D \cap \{1, \ldots, n\}$ is the least element of D. ■

Theorem 1.4 (4.2, strong induction)

Let $A \subset \mathbb{Z}_+$ be such that $\{1, \ldots, n-1\} \subset A$ implies $n \in A$, for each n. Then $A = \mathbb{Z}_+$.

Proof. Otherwise let n be the least element of $\mathbb{Z}_+ - A$; then $\{1, \ldots, n-1\} \subset A$, so $n \in A$, a contradiction. ■

The **Archimedean ordering property** ($\mathbb{Z}_+$ is unbounded above in $\mathbb{R}$) follows from the least upper bound property: if $b = \sup \mathbb{Z}_+$ existed, $b - 1 < n$ for some n, so $n + 1 > b$.

1.5 Cartesian products

For an indexed family $\{A_\alpha\}_{\alpha \in J}$, the **cartesian product** $\prod_{\alpha \in J} A_\alpha$ is the set of all functions $x : J \to \bigcup A_\alpha$ with $x(\alpha) \in A_\alpha$ (written $(x_\alpha)_{\alpha \in J}$). For finite $J = \{1, \ldots, m\}$ these are m-tuples; for $J = \mathbb{Z}_+$ they are sequences; X^J denotes the set of all functions $J \to X$, e.g. $\mathbb{R}^\omega$, the set of real sequences.

1.6 Finite sets

Motivation. "Finite" must be defined through bijections with $\{1,\dots,n\}$, and the intuitive facts (a finite set has a unique size, subsets of finite sets are finite) need proofs. The pigeonhole principle proved here is the prototype of every counting argument in the course.

A set is **finite** if it is empty or in bijection with a section $\{1,\dots,n\}$; we show n is unique.

Lemma 1.5 (6.1)

Let $a_0 \in A$. There is a bijection $A \to \{1,\dots,n+1\}$ iff there is a bijection $A - \{a_0\} \to \{1,\dots,n\}$.

Proof. Given $g : A - \{a_0\} \to \{1,\dots,n\}$, extend by $a_0 \mapsto n+1$. Conversely, given $f : A \to \{1,\dots,n+1\}$, if $f(a_0) = n+1$ restrict; otherwise compose f with the transposition of $f(a_0)$ and $n+1$ first. ∎

Theorem 1.6 (6.2)

Let $f : A \to \{1,\dots,n\}$ be a bijection and B a proper subset of A. Then there is no bijection $B \to \{1,\dots,n\}$, but (if $B \neq \emptyset$) there is a bijection $B \to \{1,\dots,m\}$ for some $m < n$.

Proof. Induction on n. For $n = 1$, A is a point and $B = \emptyset$. Assume the result for n and let $f : A \to \{1,\dots,n+1\}$, $B \subsetneq A$ nonempty. Pick $a_0 \in B$ (or, if convenient, any point); $A - \{a_0\}$ is in bijection with $\{1,\dots,n\}$ by Lemma 1.5. If $B - \{a_0\}$ is a proper subset of $A - \{a_0\}$ (true since $B \neq A$ and $a_0 \in B$), the induction hypothesis says $B - \{a_0\}$ is empty or in bijection with $\{1,\dots,p\}$, $p < n$, and not with $\{1,\dots,n\}$; by Lemma 1.5 again, B is in bijection with $\{1,\dots,p+1\}$, $p+1 < n+1$, and not with $\{1,\dots,n+1\}$. ∎

Corollary 1.7 (6.3–6.5)

A finite set has no bijection with a proper subset of itself; $\mathbb{Z}_+$ is not finite; the cardinality of a finite set is uniquely determined.

Proof. The first is Theorem 1.6. $n \mapsto n+1$ is a bijection of $\mathbb{Z}_+$ onto a proper subset. If A were in bijection with $\{1,\dots,m\}$ and $\{1,\dots,n\}$, $m < n$, then $\{1,\dots,n\}$ would be in bijection with its proper subset $\{1,\dots,m\}$. ∎

Corollary 1.8 (6.6)

A subset B of a finite set A is finite; if B is proper, its cardinality is smaller.

Corollary 1.9 (6.7)

For nonempty B the following are equivalent:
(1) B is finite;
(2) there is a surjection from some section $\{1,\dots,n\}$ onto B;
(3) there is an injection of B into some section.

Proof. (1)⇒(2) clear. (2)⇒(3): send b to the least element of $f^{-1}(b)$ (Theorem 1.3). (3)⇒(1): B is in bijection with a subset of a finite set; use Corollary 1.8. ∎

Corollary 1.10 (6.8)

Finite unions and finite cartesian products of finite sets are finite.

Proof. For $A \cup B$ with $A \cong \{1, \ldots, m\}$ and $B \cong \{1, \ldots, n\}$, the map onto $\{1, \ldots, m+n\}$ sending B to the last n values gives a surjection from a section; apply Corollary 1.9. $A \times B$ is the union of the finite sets $A \times \{b\}$; induct for more factors. ■

1.7 Countable and uncountable sets

Motivation. Many topological properties (second countability, separability) are statements about countable families. We need efficient criteria for countability, such as countable unions of countable sets, and Cantor's diagonal argument shows that not every infinite set is countable.

A set is **countably infinite** if it is in bijection with $\mathbb{Z}_+$, **countable** if finite or countably infinite, and **uncountable** otherwise.

Theorem 1.11 (7.1)

For nonempty B the following are equivalent:
(1) B is countable;
(2) there is a surjection $f : \mathbb{Z}_+ \to B$;
(3) there is an injection $g : B \to \mathbb{Z}_+$.

Proof. (1)⇒(2): if B is finite, extend a surjection from a section by a constant. (2)⇒(3): $g(b) = \min f^{-1}(b)$. (3)⇒(1): $g(B) \subset \mathbb{Z}_+$ is countable by the next lemma. ■

Lemma 1.12 (7.2)

An infinite subset $C \subset \mathbb{Z}_+$ is countably infinite.

Proof. Define $h(1) = \min C$ and $h(n) = \min\{C - h(\{1, \ldots, n-1\})\}$, which is nonempty because C is infinite (recursive definition, Theorem 1.21). h is injective (increasing). It is surjective: if $c \in C$ were not in the image, then $h(n) < c$ for all n (since $h(n)$ is the least element not yet chosen and c is available), so h would map $\mathbb{Z}_+$ injectively into $\{1, \ldots, c\}$, contradicting Corollary 1.7. ■

Corollary 1.13 (7.3)

A subset of a countable set is countable.

Corollary 1.14 (7.4)

$\mathbb{Z}_+ \times \mathbb{Z}_+$ is countably infinite.

Proof. $(n, m) \mapsto 2^n 3^m$ is injective by unique factorization; apply Theorem 1.11. ■

Theorem 1.15 (7.5)

A countable union of countable sets is countable.

Proof. Let $\{A_n\}_{n\in J}$, $J \subset \mathbb{Z}_+$, be countable and nonempty. Choose surjections $f_n : \mathbb{Z}_+ \to A_n$ (and, if J is finite, a surjection $g : \mathbb{Z}_+ \to J$). Then $(k, m) \mapsto f_{g(k)}(m)$ maps $\mathbb{Z}_+ \times \mathbb{Z}_+$ onto $\bigcup A_n$; compose with a surjection $\mathbb{Z}_+ \to \mathbb{Z}_+ \times \mathbb{Z}_+$. (The simultaneous choice of the f_n uses the axiom of choice.) ■

Theorem 1.16 (7.6)

A finite product of countable sets is countable.

Proof. $A \times B$ is the image of $\mathbb{Z}_+ \times \mathbb{Z}_+$ under $(n, m) \mapsto (f(n), g(m))$; induct on the number of factors. ■

Theorem 1.17 (7.7)

$\{0, 1\}^\omega$ is uncountable.

Proof. Given $g : \mathbb{Z}_+ \to \{0, 1\}^\omega$, write $g(n) = (x_{n1}, x_{n2}, \dots)$ and let $y_n = 1 - x_{nn}$. Then $y = (y_n)$ differs from $g(n)$ in the n-th coordinate, so g is not surjective. ■

Theorem 1.18 (7.8)

For any set A there is no injection $\mathcal{P}(A) \to A$ and no surjection $A \to \mathcal{P}(A)$.

Proof. An injection $\mathcal{P}(A) \to A$ would yield a surjection $A \to \mathcal{P}(A)$. Given $g : A \to \mathcal{P}(A)$, let $B = \{a : a \notin g(a)\}$. If $B = g(a_0)$, then $a_0 \in B \iff a_0 \notin g(a_0) = B$, a contradiction. ■

1.8 The principle of recursive definition

Motivation. Sequences are often built step by step, each term depending on the previous ones. The recursion principle justifies such definitions rigorously, and it is used every time we "choose x_1, then x_2, ...".

Fix a set C and the requirement $(*)$: $h(1) =$ the least element of C, and $h(i) =$ the least element of $C - h(\{1, \dots, i-1\})$ (with $C \subset \mathbb{Z}_+$ infinite, as in Lemma 1.12); the general case is identical.

Lemma 1.19 (8.1)

For each n there is a function $f : \{1, \dots, n\} \to C$ satisfying $(*)$ on its domain.

Proof. Induction: extend a function on $\{1, \dots, n\}$ by the value prescribed by $(*)$ at $n+1$ (the set $C - f(\{1, \dots, n\})$ is nonempty since C is infinite). ■

Lemma 1.20 (8.2)

Two functions on $\{1, \dots, n\}$ and $\{1, \dots, m\}$ satisfying $(*)$ agree on their common domain.

Proof. If not, let i be the least point of disagreement; $(*)$ determines the value at i from the (equal) values below i. ■

Theorem 1.21 (8.3–8.4, principle of recursive definition)

Let A be a set, $a_0 \in A$, and ρ a function assigning to each function f from a nonempty section $\{1, \dots, n\}$ into A an element $\rho(f) \in A$. Then there is a unique $h : \mathbb{Z}_+ \to A$ with $h(1) = a_0$ and $h(i) = \rho(h|_{\{1,\dots,i-1\}})$ for $i > 1$.

Proof. As in Lemmas 1.19 and 1.20, for each n there is a unique function on $\{1, \dots, n\}$ satisfying the formula, and these are compatible; their union is a function h on $\mathbb{Z}_+$ with the property, unique by the same argument. ■

1.9 Infinite sets and the axiom of choice

Motivation. Some infinite constructions require infinitely many arbitrary choices with no rule to make them. The axiom of choice makes this legitimate; we isolate where it is needed, for instance to show that every infinite set contains a countably infinite subset.

Theorem 1.22 (9.1)

For a set A the following are equivalent:
(1) there is an injection $\mathbb{Z}_+ \to A$;
(2) there is a bijection of A with a proper subset of itself;
(3) A is infinite.

Proof. (1)⇒(2): if $f : \mathbb{Z}_+ \to A$ is injective, define $g(f(n)) = f(n+1)$ and $g(x) = x$ off $f(\mathbb{Z}_+)$; g is a bijection onto $A - \{f(1)\}$.

(2)⇒(3) is Corollary 1.7. (3)⇒(1): let c be a choice function on the nonempty subsets of A (Lemma 1.23) and define recursively $f(1) = c(A)$, $f(n) = c(A - f(\{1, \dots, n-1\}))$; the sets are nonempty because A is infinite, and f is injective. ■

Definition (Axiom of choice)

Given a collection $\mathcal{A}$ of disjoint nonempty sets, there is a set C consisting of exactly one element from each member of $\mathcal{A}$.

Lemma 1.23 (9.2, choice function)

Given a collection $\mathcal{B}$ of nonempty sets (not necessarily disjoint), there is a function $c : \mathcal{B} \to \bigcup \mathcal{B}$ with $c(B) \in B$ for each B.

Proof. The sets $B' = \{B\} \times B$, $B \in \mathcal{B}$, are disjoint and nonempty. A set C meeting each B' in one point is a set of pairs (B, x) with exactly one pair for each B; it is the desired function. ■

1.10 Well-ordered sets

Motivation. Well-ordered sets allow induction beyond the integers. The minimal uncountable well-ordered set S_Ω built here is the source of many counterexamples later in the course.

An ordered set is **well-ordered** if every nonempty subset has a smallest element. Subsets of well-ordered sets are well-ordered, and so are dictionary-ordered finite products of well-ordered sets. For α in a well-ordered set X, the **section** is $S_\alpha = \{x : x < \alpha\}$.

Theorem 1.24 (10.1)

Every nonempty finite ordered set has the order type of a section $\{1, \dots, n\}$; hence it is well-ordered.

Proof. Induction on the cardinality: remove the largest element (which exists in a finite ordered set, by induction) and append it at $n+1$. ■

Theorem 1.25 (Well-ordering theorem, Zermelo)

Every set can be well-ordered. (Equivalent to the axiom of choice.)

We accept this; a proof is a transfinite version of Theorem 1.22 using a choice function. In particular there exist uncountable well-ordered sets.

Lemma 1.26 (10.2)

There is a well-ordered set A with a largest element Ω such that the section S_Ω is uncountable but every other section of A is countable.

Proof. Let B be an uncountable well-ordered set and $C = \{1, 2\} \times B$ in dictionary order; some section of C is uncountable (e.g. that of $2 \times b$). Let Ω be the least element of C whose section is uncountable, and $A = S_\Omega \cup \{\Omega\}$. ■

S_Ω is called the **minimal uncountable well-ordered set**, and $\overline{S}_\Omega = S_\Omega \cup \{\Omega\}$.

Theorem 1.27 (10.3)

Every countable subset A of S_Ω has an upper bound in S_Ω.

Proof. $\bigcup_{a \in A} S_a$ is a countable union of countable sets, hence countable, so it is not all of S_Ω; any x outside it is an upper bound for A. ■

1.11 The maximum principle

Motivation. Zorn's lemma is the tool for proving that maximal objects exist (maximal ideals, bases, maximal filters); it is equivalent to the axiom of choice and will be used in the proof of the Tychonoff theorem.

A **strict partial order** $\prec$ is nonreflexive and transitive; a subset on which it is a simple order is a **chain**.

Theorem 1.28 (11.1, Hausdorff maximum principle)

In a set with a strict partial order there is a maximal chain (a chain not properly contained in any other chain).

Proof. Well-order the set A by $<$ and define by transfinite recursion $h(\alpha) = \mathit{true}$ iff $\{\beta < \alpha : h(\beta) = \mathit{true}\} \cup \{\alpha\}$ is a chain for $\prec$. The set B of α with $h(\alpha)$ true is a chain (any two of its elements lie in the chain tested at the later one). It is maximal: if $B \cup \{\alpha\}$ were a chain for some $\alpha \notin B$, then $\{\beta < \alpha : \beta \in B\} \cup \{\alpha\}$ would be a chain, forcing $\alpha \in B$. ■

Theorem 1.29 (Zorn's lemma)

If A is strictly partially ordered and every chain has an upper bound in A, then A has a maximal element.

Proof. Take a maximal chain B and an upper bound c of B. If $c \prec d$ for some d, then $B \cup \{d\}$ is a chain properly containing B (note $d \notin B$ since $d \succ c \succeq$ every element of B), contradicting maximality. ■

Example 1.1

Every vector space has a basis: the linearly independent sets, ordered by proper inclusion, satisfy the hypothesis of Zorn's lemma (a union of a chain is independent), and a maximal independent set spans. Zorn's lemma, the well-ordering theorem, the Hausdorff maximum principle and the axiom of choice are all equivalent.

Chapter 2. Topological Spaces and Continuous Functions

2.1 Topological spaces

Motivation. Continuity, convergence and connectedness in $\mathbb{R}^n$ can all be described using only open sets, without any distances. A topology keeps exactly this information, and the definition below lists the properties of open sets in $\mathbb{R}^n$ that the proofs actually use.

Definition (Topology)

A **topology** on a set X is a collection $\mathcal{T}$ of subsets of X, called **open sets**, such that $\emptyset, X \in \mathcal{T}$, arbitrary unions of members of $\mathcal{T}$ are in $\mathcal{T}$, and finite intersections of members of $\mathcal{T}$ are in $\mathcal{T}$.

Examples: the **discrete topology** $\mathcal{P}(X)$; the **trivial topology** $\{\emptyset, X\}$; the **finite complement topology** $\mathcal{T}_f = \{U : X - U \text{ finite}\} \cup \{\emptyset\}$ (a topology because $X - \bigcup U_\alpha = \bigcap (X - U_\alpha)$ and $X - \bigcap_{i \le n} U_i = \bigcup_{i \le n} (X - U_i)$ are finite); the countable complement topology. If $\mathcal{T} \subset \mathcal{T}'$ then $\mathcal{T}'$ is **finer** and $\mathcal{T}$ **coarser**.

2.2 Basis for a topology

Motivation. Open sets in $\mathbb{R}^2$ are complicated, but each is a union of open discs. A basis is a small collection like the discs from which all open sets are generated; most topologies are specified this way, and most verifications only need to be checked on basis elements.

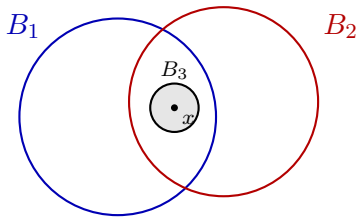


Figure. *The second basis axiom: if $x \in B_1 \cap B_2$, some basis element B_3 with $x \in B_3 \subset B_1 \cap B_2$ exists. This is what makes finite intersections of open sets open.*

Definition (Basis)

A collection $\mathcal{B}$ of subsets of X is a **basis** if (1) each x lies in at least one $B \in \mathcal{B}$, and (2) if $x \in B_1 \cap B_2$ then some $B_3 \in \mathcal{B}$ has $x \in B_3 \subset B_1 \cap B_2$. The topology **generated** by $\mathcal{B}$: U is open iff for each $x \in U$ there is $B \in \mathcal{B}$ with $x \in B \subset U$.

That this is a topology: unions are clear; for $U_1 \cap U_2$ and x in it, pick $B_i \ni x$ inside U_i and then $B_3 \subset B_1 \cap B_2$; finite intersections follow by induction. Examples: circular regions or rectangles in $\mathbb{R}^2$; one-point sets (discrete topology).

Lemma 2.1 (13.1)

If $\mathcal{B}$ is a basis for $\mathcal{T}$, then $\mathcal{T}$ equals the collection of all unions of elements of $\mathcal{B}$.

Proof. Unions of basis elements are open. An open U is the union of the basis elements B_x with $x \in B_x \subset U$, $x \in U$. ■

Lemma 2.2 (13.2)

Let $\mathcal{C}$ be a collection of open sets of X such that for each open U and $x \in U$ there is $C \in \mathcal{C}$ with $x \in C \subset U$. Then $\mathcal{C}$ is a basis for the topology of X.

Proof. Condition (1) holds with $U = X$; for (2) apply the hypothesis to the open set $C_1 \cap C_2$. The topology $\mathcal{T}'$ generated by $\mathcal{C}$ contains $\mathcal{T}$ by hypothesis, and is contained in $\mathcal{T}$ because its sets are unions of members of $\mathcal{C} \subset \mathcal{T}$. ■

Lemma 2.3 (13.3)

Let $\mathcal{B}, \mathcal{B}'$ be bases for $\mathcal{T}, \mathcal{T}'$ on X. Then $\mathcal{T}'$ is finer than $\mathcal{T}$ iff for each x and each $B \in \mathcal{B}$ containing x there is $B' \in \mathcal{B}'$ with $x \in B' \subset B$.

Proof. ($\Leftarrow$) If $U \in \mathcal{T}$ and $x \in U$, choose B with $x \in B \subset U$, then B'; so $U \in \mathcal{T}'$.

($\Rightarrow$) $B \in \mathcal{T} \subset \mathcal{T}'$, and $\mathcal{T}'$ is generated by $\mathcal{B}'$. ■

On $\mathbb{R}$: the **standard topology** (basis: open intervals); the **lower limit topology** $\mathbb{R}_\ell$ (basis: $[a, b)$); the **K-topology** $\mathbb{R}_K$ (basis: (a, b) and $(a, b) - K$, $K = \{1/n : n \in \mathbb{Z}_+\}$).

Lemma 2.4 (13.4)

The topologies of $\mathbb{R}_\ell$ and $\mathbb{R}_K$ are strictly finer than the standard topology, but not comparable with one another.

Proof. Given $(a, b) \ni x$, both $[x, b)$ and (a, b) lie inside it, so both are finer (Lemma 2.3). $[x, d)$ contains no open interval around x, and $(-1, 1) - K$ contains no open interval around 0, so both are strictly finer. No basis element of $\mathbb{R}_K$ containing 10 lies in $[10, 11)$, and no basis element of $\mathbb{R}_\ell$ containing 0 lies in $(-1, 1) - K$. ■

A **subbasis** $\mathcal{S}$ is a collection of subsets whose union is X; the topology it generates is the collection of unions of finite intersections of members of $\mathcal{S}$ (these finite intersections form a basis).

2.3 The order topology

Motivation. Any linearly ordered set, not just $\mathbb{R}$, carries a topology generated by open intervals. This gives a supply of examples, such as the ordered square, that test our intuition.

Let X be simply ordered with more than one element. The **order topology** has as basis: all open intervals (a, b); all $[a_0, b)$ if X has a smallest element a_0; all $(a, b_0]$ if X has a largest element b_0. (If $x \in B_1 \cap B_2$, the intersection of two such sets is again of this form.) The open rays $(a, +\infty)$, $(-\infty, a)$ form a subbasis. Examples: $\mathbb{R}$ (standard topology); $\mathbb{Z}_+$ (discrete); $\mathbb{R} \times \mathbb{R}$ in the dictionary order (basis: vertical intervals $(a \times b, a \times d)$); $\{1, 2\} \times \mathbb{Z}_+$ in the dictionary order, which is not discrete since every neighbourhood of 2×1 contains points $1 \times n$.

2.4 The product topology on $X \times Y$

Motivation. To treat $\mathbb{R}^2$ as built from two copies of $\mathbb{R}$, we need to topologize $X \times Y$ from the topologies of X and Y; products of open sets form a basis.

The **product topology** on $X \times Y$ has basis $\mathcal{B} = \{U \times V : U \text{ open in } X,\ V \text{ open in } Y\}$ (a basis since $(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2)$).

Theorem 2.5 (15.1)

If $\mathcal{B}$ and $\mathcal{C}$ are bases for X and Y, then $\{B \times C : B \in \mathcal{B}, C \in \mathcal{C}\}$ is a basis for the topology of $X \times Y$.

Proof. Given open W and $x \times y \in W$, choose $U \times V$ with $x \times y \in U \times V \subset W$, then $B \subset U$, $C \subset V$ basis elements containing x, y. Apply Lemma 2.2. ■

Theorem 2.6 (15.2)

$\mathcal{S} = \{\pi_1^{-1}(U) : U \text{ open in } X\} \cup \{\pi_2^{-1}(V) : V \text{ open in } Y\}$ is a subbasis for the product topology.

Proof. $\pi_1^{-1}(U) \cap \pi_2^{-1}(V) = U \times V$, so the finite intersections of members of $\mathcal{S}$ are exactly the basis sets $U \times V$ (together with sets of the same form). ■

2.5 The subspace topology

Motivation. A subset of a space, such as a circle in the plane, should inherit a topology from its surroundings. The subspace topology does this, and the main subtlety is that "open" depends on which space one is in.

For $Y \subset X$, the **subspace topology** is $\mathcal{T}_Y = \{Y \cap U : U \in \mathcal{T}\}$.

Lemma 2.7 (16.1)

If $\mathcal{B}$ is a basis for X, then $\{B \cap Y : B \in \mathcal{B}\}$ is a basis for the subspace topology on Y.

Proof. For U open in X and $y \in U \cap Y$, choose B with $y \in B \subset U$; then $y \in B \cap Y \subset U \cap Y$. Apply Lemma 2.2. ■

Lemma 2.8 (16.2)

If U is open in Y and Y is open in X, then U is open in X.

Proof. $U = Y \cap V$ with V open in X, an intersection of two open sets of X. ■

Theorem 2.9 (16.3)

If A is a subspace of X and B of Y, the product topology on $A \times B$ equals the subspace topology from $X \times Y$.

Proof. $(U \times V) \cap (A \times B) = (U \cap A) \times (V \cap B)$; the left sides form a basis for the subspace topology (Lemma 2.7) and the right sides a basis for the product topology (Theorem 2.5). ■

Theorem 2.10 (16.4)

If X is ordered with the order topology and $Y \subset X$ is convex (for $a < b$ in Y, $(a, b) \subset Y$), then the order topology on Y equals the subspace topology.

Proof. The intersection of an open ray $(a, +\infty)$ of X with Y is Y, $\emptyset$, or an open ray of Y (convexity is used when $a \notin Y$: then a is a lower or upper bound of Y). So subbasis elements of the subspace topology are open in the order topology. Conversely, an open ray of Y is the intersection of Y with the corresponding ray of X. ■

Without convexity the two can differ: in $Y = [0, 1) \cup \{2\} \subset \mathbb{R}$, $\{2\}$ is open in the subspace topology but not in the order topology; in $I \times I$ with the dictionary order, the order topology differs from the subspace topology of $\mathbb{R} \times \mathbb{R}$ (dictionary order). We always give subsets of ordered sets the subspace topology unless said otherwise.

2.6 Closed sets and limit points

Motivation. Closed sets, closures and limit points describe "what can be approximated". Unlike in $\mathbb{R}^n$, a sequence in a general space may converge to two different points; the Hausdorff condition restores uniqueness of limits, and it holds in every space we care about.

A is **closed** if $X - A$ is open.

Theorem 2.11 (17.1)

$\emptyset$ and X are closed; arbitrary intersections and finite unions of closed sets are closed.

Proof. De Morgan's laws applied to the axioms for open sets. ■

Theorem 2.12 (17.2)

For a subspace Y, A is closed in Y iff $A = C \cap Y$ for some closed $C \subset X$.

Proof. If $A = C \cap Y$, then $Y - A = (X - C) \cap Y$ is open in Y. Conversely, $Y - A = U \cap Y$ with U open, and $A = (X - U) \cap Y$. ■

Theorem 2.13 (17.3)

If A is closed in Y and Y is closed in X, then A is closed in X.

The **interior** $\operatorname{Int} A$ is the union of the open sets contained in A; the **closure** $\overline{A}$ is the intersection of the closed sets containing A. So $\operatorname{Int} A \subset A \subset \overline{A}$, A is open iff $A = \operatorname{Int} A$, closed iff $A = \overline{A}$.

Theorem 2.14 (17.4)

If Y is a subspace of X, $A \subset Y$, and $\overline{A}$ is the closure of A in X, then the closure of A in Y is $\overline{A} \cap Y$.

Proof. $\overline{A} \cap Y$ is closed in Y and contains A, so it contains the closure B of A in Y. Conversely $B = C \cap Y$ with C closed in X (Theorem 2.12) and $C \supset A$, so $\overline{A} \subset C$ and $\overline{A} \cap Y \subset B$. ■

We say U is a **neighbourhood** of x if U is open and contains x, and that U **intersects** A if $U \cap A \neq \emptyset$.

Theorem 2.15 (17.5)

(a) $x \in \overline{A}$ iff every open set containing x intersects A.
(b) If the topology is given by a basis, $x \in \overline{A}$ iff every basis element containing x intersects A.

Proof. (a) Contrapositive: $x \notin \overline{A}$ iff some open $U \ni x$ misses A (take $U = X - \overline{A}$; conversely $X - U$ is a closed set containing A and missing x).

(b) Every open set containing x contains a basis element containing x. ■

For example, in $\mathbb{R}$ the closure of $(0, 1]$ is $[0, 1]$, of $\mathbb{Q}$ is $\mathbb{R}$, of $\{1/n\}$ is $\{0\} \cup \{1/n\}$. The point x is a **limit point** of A if every neighbourhood of x intersects A in a point other than x; equivalently $x \in \overline{A - \{x\}}$. Let A' denote the set of limit points.

Theorem 2.16 (17.6)

$\overline{A} = A \cup A'$.

Proof. $A' \subset \overline{A}$ by Theorem 2.15. If $x \in \overline{A} - A$, every neighbourhood of x meets A in a point, which is not x; so $x \in A'$. ■

Corollary 2.17 (17.7)

A subset is closed iff it contains all its limit points.

A sequence (x_n) **converges** to x if for every neighbourhood U of x there is N with $x_n \in U$ for $n \geq N$. In general limits need not be unique. X is a **Hausdorff space** if distinct points have disjoint neighbourhoods.

Theorem 2.18 (17.8)

Every finite point set in a Hausdorff space is closed.

Proof. For a single point x_0: any $x \neq x_0$ has a neighbourhood missing a neighbourhood of x_0, in particular missing x_0, so $x \notin \overline{\{x_0\}}$. Finite unions of closed sets are closed. ■

The condition that finite sets are closed is the T_1 **axiom**; it is weaker than Hausdorff (finite complement topology on $\mathbb{R}$).

Theorem 2.19 (17.9)

In a T_1 space, x is a limit point of A iff every neighbourhood of x contains infinitely many points of A.

Proof. ($\Rightarrow$) If a neighbourhood U of x met A in finitely many points, and $x_1, \ldots, x_m$ are those different from x, then $U - \{x_1, \ldots, x_m\}$ is a neighbourhood of x (finite sets are closed) meeting $A - \{x\}$ not at all.

($\Leftarrow$) is clear. ■

Theorem 2.20 (17.10)

In a Hausdorff space a sequence converges to at most one point.

Proof. If $x_n \to x$ and $y \neq x$, take disjoint neighbourhoods $U \ni x$, $V \ni y$; U contains all but finitely many x_n, so V cannot, and $x_n \not\to y$. ■

Theorem 2.21 (17.11)

Every simply ordered set is Hausdorff in the order topology; products of two Hausdorff spaces and subspaces of Hausdorff spaces are Hausdorff.

Proof. For $a < b$ in an ordered set: if some c lies between them, $(-\infty, c)$ and $(c, +\infty)$ separate; otherwise $(-\infty, b)$ and $(a, +\infty)$ do. For products use $U_1 \times V_1$ and $U_2 \times V_2$ (separating in one coordinate suffices); for subspaces intersect with the subspace. ■

2.7 Continuous functions

Motivation. Continuity becomes "preimages of open sets are open". This definition is simple to work with, and the results below give the standard ways to build continuous maps: restriction, composition, pasting, and maps into products.

$f : X \to Y$ is **continuous** if $f^{-1}(V)$ is open for every open $V \subset Y$; it suffices to check V in a basis or subbasis, since preimages preserve unions and intersections.

Theorem 2.22 (18.1)

For $f : X \to Y$ the following are equivalent:
(1) f is continuous;
(2) $f(\overline{A}) \subset \overline{f(A)}$ for every $A \subset X$;
(3) $f^{-1}(B)$ is closed for every closed $B \subset Y$;
(4) for each x and each neighbourhood V of $f(x)$ there is a neighbourhood U of x with $f(U) \subset V$.

Proof. (1)⇒(2): for $x \in \overline{A}$ and a neighbourhood V of $f(x)$, $f^{-1}(V)$ is a neighbourhood of x meeting A in some y, and $f(y) \in V \cap f(A)$. (2)⇒(3): with $A = f^{-1}(B)$, $f(\overline{A}) \subset \overline{f(A)} \subset \overline{B} = B$, so $\overline{A} \subset A$. (3)⇒(1): complements. (1)⇒(4): $U = f^{-1}(V)$. (4)⇒(1): $f^{-1}(V)$ is the union of the neighbourhoods U_x given by (4). ■

A **homeomorphism** is a continuous bijection with continuous inverse; equivalently a bijection with $f(U)$ open iff U open. A continuous injection $f : X \to Y$ that is a homeomorphism onto $f(X)$ (subspace topology) is an **imbedding**. Examples: the identity $\mathbb{R}_\ell \to \mathbb{R}$ is a continuous bijection but not a homeomorphism; $[0, 1) \to S^1$, $t \mapsto (\cos 2\pi t, \sin 2\pi t)$ is a continuous bijection that is not a homeomorphism.

Theorem 2.23 (18.2, rules for constructing continuous functions)

(a) Constant functions are continuous.
(b) Inclusions of subspaces are continuous.
(c) Composites of continuous functions are continuous.
(d) Restricting the domain of a continuous function to a subspace keeps it continuous.
(e) Restricting or expanding the range: if $f : X \to Y$ is continuous and $Z \subset Y$ contains $f(X)$, then $f : X \to Z$ is continuous; if $Y \subset Z$ is a subspace, $f : X \to Z$ is continuous.
(f) Local formulation: f is continuous if X is a union of open sets U_α with each $f|_{U_\alpha}$ continuous.

Proof. (a) Preimages are $\emptyset$ or X.

(b) $j^{-1}(U) = U \cap A$.

(c) $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W))$.

(d) Composite with the inclusion.

(e) $f^{-1}(Z \cap V) = f^{-1}(V)$; and composite with the inclusion $Y \to Z$.

(f) $f^{-1}(V) = \bigcup (f|_{U_\alpha})^{-1}(V)$, a union of sets open in U_α, hence open in X (Lemma 2.8). ■

Theorem 2.24 (18.3, the pasting lemma)

Let $X = A \cup B$ with A, B closed, $f : A \to Y$ and $g : B \to Y$ continuous and $f = g$ on $A \cap B$. Then $h = f \cup g : X \to Y$ is continuous. (The same holds with A, B open.)

Proof. For closed $C \subset Y$, $h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$; each is closed in A or B, hence in X (Theorem 2.13). ■

Theorem 2.25 (18.4, maps into products)

$f : A \to X \times Y$, $f = (f_1, f_2)$, is continuous iff f_1 and f_2 are continuous.

Proof. The projections are continuous ($\pi_1^{-1}(U) = U \times Y$), so $f_i = \pi_i \circ f$ are continuous if f is. Conversely $f^{-1}(U \times V) = f_1^{-1}(U) \cap f_2^{-1}(V)$. ■

2.8 The product topology

Motivation. For infinite products there are two natural candidates, the box and product topologies. The product topology is the right one: it is the one for which a map into the product is continuous exactly when each coordinate is, and it is the one for which Tychonoff's theorem holds.

For an indexed family $\{X_\alpha\}_{\alpha \in J}$, the **box topology** on $\prod X_\alpha$ has basis all products $\prod U_\alpha$ of open sets, and the **product topology** has subbasis $\mathcal{S} = \bigcup_\beta \{\pi_\beta^{-1}(U_\beta)\}$.

Theorem 2.26 (19.1, comparison)

The box topology has basis all $\prod U_\alpha$ with U_α open; the product topology has basis all $\prod U_\alpha$ with U_α open and $U_\alpha = X_\alpha$ for all but finitely many α. For finite products they coincide; in general the box topology is finer.

Proof. A finite intersection $\pi_{\beta_1}^{-1}(U_{\beta_1}) \cap \cdots \cap \pi_{\beta_n}^{-1}(U_{\beta_n})$ is exactly a product $\prod U_\alpha$ with $U_\alpha = X_\alpha$ except for $\alpha \in \{\beta_1, \ldots, \beta_n\}$. ■

Theorem 2.27 (19.2)

If each X_α has a basis $\mathcal{B}_\alpha$, products $\prod B_\alpha$ with $B_\alpha \in \mathcal{B}_\alpha$ form a basis for the box topology, and those with $B_\alpha \in \mathcal{B}_\alpha$ for finitely many α and $B_\alpha = X_\alpha$ otherwise form a basis for the product topology.

Theorem 2.28 (19.3)

If $A_\alpha \subset X_\alpha$, then $\prod A_\alpha$ is a subspace of $\prod X_\alpha$ when both carry the box topology, or both the product topology.

Theorem 2.29 (19.4)

A product of Hausdorff spaces is Hausdorff in both topologies.

Theorem 2.30 (19.5)

In either topology, $\prod \overline{A_\alpha} = \overline{\prod A_\alpha}$.

Proof. If $x \in \prod \overline{A_\alpha}$ and $U = \prod U_\alpha$ is a basis element containing x, choose $y_\alpha \in U_\alpha \cap A_\alpha$; then $y \in U \cap \prod A_\alpha$. Conversely, if $x \in \overline{\prod A_\alpha}$ and $V_\beta \ni x_\beta$ is open, $\pi_\beta^{-1}(V_\beta)$ meets $\prod A_\alpha$, so V_β meets A_β. (Theorems 2.27–2.29 are proved like their two-factor versions.) ■

Theorem 2.31 (19.6)

Let $f : A \to \prod X_\alpha$, $f(a) = (f_\alpha(a))$, where $\prod X_\alpha$ has the product topology. Then f is continuous iff each f_α is continuous.

Proof. π_β is continuous, giving necessity. Conversely, $f^{-1}(\pi_\beta^{-1}(U_\beta)) = f_\beta^{-1}(U_\beta)$ is open, and these sets form a subbasis. ■

This fails for the box topology: $f : \mathbb{R} \to \mathbb{R}^\omega$, $f(t) = (t, t, \dots)$, has continuous coordinates, but $f^{-1}(\prod(-\frac{1}{n}, \frac{1}{n})) = \{0\}$. Henceforth products carry the product topology.

2.9 The metric topology

Motivation. Metric spaces are the motivating examples of topological spaces. Here we check that the familiar metrics on $\mathbb{R}^n$ give the product topology, and that the uniform metric on $\mathbb{R}^J$ lies strictly between the product and box topologies.

A **metric** d on X satisfies $d(x, y) \geq 0$ with equality iff $x = y$, symmetry, and the triangle inequality. The ϵ-balls $B_d(x, \epsilon)$ form a basis (if $y \in B_d(x, \epsilon)$, then $B_d(y, \epsilon - d(x, y)) \subset B_d(x, \epsilon)$), generating the **metric topology**; U is open iff each $y \in U$ has a ball $B_d(y, \delta) \subset U$. X is **metrizable** if some metric induces its topology. A is **bounded** if $\sup_{a,b \in A} d(a, b) = \operatorname{diam} A < \infty$.

Theorem 2.32 (20.1)

$\bar{d}(x, y) = \min\{d(x, y), 1\}$ is a metric inducing the same topology as d (the **standard bounded metric**).

Proof. Triangle inequality: if $d(x, y) \geq 1$ or $d(y, z) \geq 1$ the right side is at least 1; otherwise $\bar{d}(x, z) \leq d(x, z) \leq d(x, y) + d(y, z) = \bar{d}(x, y) + \bar{d}(y, z)$. The balls of radius < 1 coincide and form a basis for both topologies. ■

Lemma 2.33 (20.2)

For metrics d, d' on X with topologies $\mathcal{T}, \mathcal{T}'$, $\mathcal{T}'$ is finer than $\mathcal{T}$ iff for each x and $\epsilon > 0$ there is $\delta > 0$ with $B_{d'}(x, \delta) \subset B_d(x, \epsilon)$.

Proof. This is Lemma 2.3 for the bases of balls, using that every ball containing x contains a ball centred at x. ■

Theorem 2.34 (20.3)

The euclidean metric d and the square metric $\rho(x, y) = \max|x_i - y_i|$ on $\mathbb{R}^n$ both induce the product topology.

Proof. $\rho \le d \le \sqrt{n}\rho$ gives $B_d(x, \epsilon) \subset B_\rho(x, \epsilon)$ and $B_\rho(x, \epsilon/\sqrt{n}) \subset B_d(x, \epsilon)$, so they agree. Product basis elements $\prod(a_i, b_i) \ni x$ contain $B_\rho(x, \epsilon)$ for $\epsilon = \min_i \min(x_i - a_i, b_i - x_i)$, and $B_\rho(x, \epsilon)$ is itself a product of intervals. ■

On $\mathbb{R}^J$ the **uniform metric** is $\bar{\rho}(x, y) = \sup_\alpha \bar{d}(x_\alpha, y_\alpha)$.

Theorem 2.35 (20.4)

The uniform topology on $\mathbb{R}^J$ is finer than the product topology and coarser than the box topology; if J is infinite, all three are different.

Proof. A product basis element $\prod U_\alpha$ restricting $\alpha_1, \dots, \alpha_n$ contains a ball $B_{\bar{\rho}}(x, \epsilon)$ (ϵ the minimum of the radii needed in those coordinates, and < 1). A ball $B_{\bar{\rho}}(x, \epsilon)$ contains the box set $\prod(x_\alpha - \epsilon/2, x_\alpha + \epsilon/2)$. For infinite J: $B_{\bar{\rho}}(0, 1)$ is not product-open, and $\prod(-1/n, 1/n)$ (indexing a countable subset) contains no uniform ball around 0. ■

Theorem 2.36 (20.5)

With $\bar{d}(a, b) = \min\{|a - b|, 1\}$, $D(x, y) = \sup_i \bar{d}(x_i, y_i)/i$ is a metric inducing the product topology on $\mathbb{R}^\omega$.

Proof. D is a metric (sup of metrics divided by i). Given U open in the product topology and $x \in U$, choose a basis set $\prod_{i \le N}(x_i - \epsilon, x_i + \epsilon) \times \mathbb{R} \times \cdots \subset U$, $\epsilon < 1$; if $D(x, y) < \epsilon/N$ then $\bar{d}(x_i, y_i) < \epsilon$ for $i \le N$, so $B_D(x, \epsilon/N) \subset U$. Conversely, given $B_D(x, \epsilon)$, choose N with $1/N < \epsilon$ and let $V = \prod_{i<N}(x_i - \epsilon/2, x_i + \epsilon/2) \times \mathbb{R} \times \mathbb{R} \times \cdots$. For $y \in V$, $\bar{d}(x_i, y_i)/i \le \epsilon/2$ for $i < N$ and $\le 1/N$ for $i \ge N$, so $D(x, y) \le \max(\epsilon/2, 1/N) < \epsilon$ and $V \subset B_D(x, \epsilon)$. ■

2.10 The metric topology (continued)

Motivation. In metric spaces, sequences suffice to detect closure and continuity, and uniform limits of continuous functions are continuous. These facts fail in general spaces, so it matters that we know exactly where metrizability is used.

Subspaces of metric spaces are metric spaces (restrict d); order topologies may or may not be metrizable; Hausdorff is necessary for metrizability.

Theorem 2.37 (21.1)

For metrizable X, Y with metrics d_X, d_Y, $f : X \to Y$ is continuous iff for each x and $\epsilon > 0$ there is $\delta > 0$ with $d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \epsilon$.

Proof. Apply Theorem 2.22(4) with neighbourhoods taken to be balls. ■

Lemma 2.38 (21.2, the sequence lemma)

If a sequence of points of A converges to x, then $x \in \overline{A}$; the converse holds if X is metrizable (indeed if X has a countable neighbourhood basis at x).

Proof. Every neighbourhood of x contains points of the sequence. Conversely, if $x \in \overline{A}$ and X is metrizable, choose $x_n \in B(x, 1/n) \cap A$; then $x_n \to x$. ■

Theorem 2.39 (21.3)

If f is continuous, then $x_n \to x$ implies $f(x_n) \to f(x)$; the converse holds if X is metrizable.

Proof. If $V \ni f(x)$ is open, $f^{-1}(V) \ni x$ contains x_n for large n. Conversely, for metrizable X we show $f(\overline{A}) \subset \overline{f(A)}$: if $x \in \overline{A}$, choose $x_n \in A$ with $x_n \to x$ (Lemma 2.38); then $f(x_n) \to f(x)$, so $f(x) \in \overline{f(A)}$. Apply Theorem 2.22. ■

In particular $\mathbb{R}^\omega$ with the box topology and $\mathbb{R}^J$ (J uncountable) in the product topology are not metrizable: in both, a point lies in the closure of a set without being a limit of a sequence from it.

Lemma 2.40 (21.4)

Addition, subtraction and multiplication $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$ and division $\mathbb{R} \times (\mathbb{R} - \{0\}) \to \mathbb{R}$ are continuous.

Proof. Standard ϵ–δ estimates using the metric $|a - a'| + |b - b'|$ on $\mathbb{R}^2$; e.g. $|ab - a'b'| \le |a||b - b'| + |b'||a - a'|$. ■

Theorem 2.41 (21.5)

If $f, g : X \to \mathbb{R}$ are continuous, so are $f \pm g$, fg, and f/g where $g \neq 0$.

Proof. $f + g$ is the composite of $(f, g) : X \to \mathbb{R} \times \mathbb{R}$ (Theorem 2.25) and addition (Lemma 2.40). ■

$f_n : X \to Y$ (Y metric) **converges uniformly** to f if for every ϵ there is N with $d(f_n(x), f(x)) < \epsilon$ for all $n \ge N$ and all x.

Theorem 2.42 (21.6, uniform limit theorem)

If continuous $f_n : X \to Y$ converge uniformly to f, then f is continuous.

Proof. Let V be open in Y, $x_0 \in f^{-1}(V)$, and $B(f(x_0), \epsilon) \subset V$. Choose N with $d(f_N(x), f(x)) < \epsilon/3$ for all x, and a neighbourhood U of x_0 with $f_N(U) \subset B(f_N(x_0), \epsilon/3)$. For $x \in U$, $d(f(x), f(x_0)) < \epsilon$, so $f(U) \subset V$. ■

2.11 The quotient topology

Motivation. Gluing the ends of an interval gives a circle; gluing sides of a square gives a torus. The quotient topology formalizes gluing, and its universal property tells us when a map defined on the glued space is continuous.

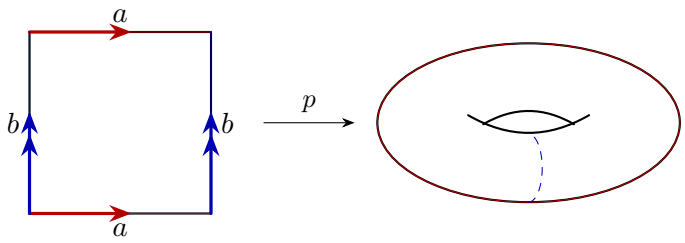


Figure. *Gluing opposite sides of a square (identify the edges labelled a, and those labelled b, respecting arrows) gives the torus; the quotient map p carries the quotient topology.*

A surjective map $p : X \to Y$ is a **quotient map** if $U \subset Y$ is open iff $p^{-1}(U)$ is open. Equivalently, p is continuous and maps **saturated** open sets (sets of the form $p^{-1}(V)$) to open sets. Open or closed continuous surjections are quotient maps (e.g. the projection $\pi_1 : X \times Y \to X$ is open). If X is a space and $p : X \to A$ a surjection onto a set, the **quotient topology** on A is the unique topology making p a quotient map; for a partition X^* of X we get the **quotient space** X^*.

Theorem 2.43 (22.1)

Let $p : X \to Y$ be a quotient map, $A \subset X$ saturated, and $q = p|_A : A \to p(A)$.
(1) If A is open or closed, q is a quotient map.
(2) If p is open or closed, q is a quotient map.

Proof. For $V \subset p(A)$, saturation gives $q^{-1}(V) = p^{-1}(V)$ and $p(U \cap A) = p(U) \cap p(A)$ for U open (or closed).

(1) If A is open and $q^{-1}(V)$ is open in A, it is open in X, so $p^{-1}(V)$ is open and V is open in Y, hence in $p(A)$.

(2) If p is open and $q^{-1}(V) = U \cap A$ with U open, then $V = q(U \cap A) = p(U) \cap p(A)$ is open in $p(A)$. Closed cases are the same with closed sets. ■

Composites of quotient maps are quotient maps; products of quotient maps need not be; quotients of Hausdorff spaces need not be Hausdorff.

Theorem 2.44 (22.2)

Let $p : X \to Y$ be a quotient map and $g : X \to Z$ constant on each fibre $p^{-1}(y)$. Then g induces $f : Y \to Z$ with $f \circ p = g$; f is continuous iff g is, and f is a quotient map iff g is.

Proof. f is well defined. For $V \subset Z$ open, $g^{-1}(V) = p^{-1}(f^{-1}(V))$ is open iff $f^{-1}(V)$ is open (quotient map). If g is a quotient map, f is surjective, and $f^{-1}(V)$ open implies $g^{-1}(V)$ open, hence V open; conversely composites of quotient maps are quotient maps. ■

Corollary 2.45 (22.3)

Let $g : X \to Z$ be a continuous surjection and $X^* = \{g^{-1}(z)\}$ with the quotient topology.
(a) g induces a continuous bijection $f : X^* \to Z$, which is a homeomorphism iff g is a quotient

map.
(b) If Z is Hausdorff, so is X^*.

Proof. (a) is Theorem 2.44.

(b) Disjoint open sets in Z pull back by the continuous f to disjoint open sets in X^*. ■

Example 2.1

(a) Collapsing ∂B^2 to a point gives S^2; identifying opposite edges of the square gives the torus.
(b) $p : \mathbb{R} \to S^1$, $p(t) = (\cos 2\pi t, \sin 2\pi t)$ is a quotient map (it is open), so $\mathbb{R}/\mathbb{Z} \cong S^1$ by Corollary 2.45.
(c) For $\mathbb{Q} \subset \mathbb{R}$ collapsed to a point, $p \times \mathrm{id}_{\mathbb{Q}} : \mathbb{R} \times \mathbb{Q} \to \mathbb{R}^* \times \mathbb{Q}$ is not a quotient map (Munkres Example 22.7).

Chapter 3. Connectedness and Compactness

3.1 Connected spaces

Motivation. The intermediate value theorem is really a statement about the shape of $[a,b]$: it cannot be split into two disjoint open pieces. Connectedness abstracts this, and continuous images of connected spaces are connected, which gives the intermediate value theorem in any setting.

A **separation** of X is a pair U, V of disjoint nonempty open sets with $X = U \cup V$. X is **connected** if no separation exists; equivalently, the only sets both open and closed are $\emptyset$ and X.

Lemma 3.1 (23.1)

If $Y \subset X$, a separation of Y is a pair of disjoint nonempty sets A, B with $A \cup B = Y$, neither containing a limit point of the other.

Proof. If A, B separate Y, A is closed in Y, so $A = \overline{A} \cap Y$ (Theorem 2.14) and $\overline{A} \cap B = \emptyset$: B contains no limit point of A; symmetrically. Conversely, if $\overline{A} \cap B = \emptyset = A \cap \overline{B}$, then $\overline{A} \cap Y = A$ and $\overline{B} \cap Y = B$ are closed in Y, so their complements B, A are open in Y. ■

Examples: the indiscrete topology is connected; $\mathbb{Q}$ is totally disconnected (its only connected subspaces are points); $Y = [-1, 0) \cup (0, 1]$ is disconnected.

Lemma 3.2 (23.2)

If C, D separate X and $Y \subset X$ is connected, then $Y \subset C$ or $Y \subset D$.

Proof. $C \cap Y$ and $D \cap Y$ are open in Y and disjoint with union Y; if both were nonempty they would separate Y. ■

Theorem 3.3 (23.3)

The union of a collection of connected subspaces of X with a point in common is connected.

Proof. Let p be the common point and suppose C, D separate $Y = \bigcup A_\alpha$, with $p \in C$. Each A_α lies in C or D (Lemma 3.2), hence in C as $p \in A_\alpha$. So $D = \emptyset$, a contradiction. ■

Theorem 3.4 (23.4)

If A is connected and $A \subset B \subset \overline{A}$, then B is connected.

Proof. If C, D separate B, then $A \subset C$, say (Lemma 3.2); then $\overline{A} \subset \overline{C}$, and $\overline{C} \cap D = \emptyset$ (Lemma 3.1), so $B \cap D \subset \overline{A} \cap D = \emptyset$. ■

Theorem 3.5 (23.5)

The image of a connected space under a continuous map is connected.

Proof. If A, B separate $f(X)$, then $f^{-1}(A), f^{-1}(B)$ separate X. ■

Theorem 3.6 (23.6)

A finite cartesian product of connected spaces is connected.

Proof. For $X \times Y$ fix $a \times b$. For each x, $T_x = (X \times b) \cup (x \times Y)$ is connected (Theorem 3.3, with common point $x \times b$). All T_x contain $a \times b$, and their union is $X \times Y$. Induct for more factors. ■

Arbitrary products are connected in the product topology (the set of points differing from a fixed point in finitely many coordinates is a connected dense subset), but $\mathbb{R}^\omega$ in the box topology is not: bounded and unbounded sequences form a separation.

3.2 Connected subspaces of the real line

Motivation. The real line is connected because of the least upper bound property; from this we get that intervals are connected and, through path connectedness, that $\mathbb{R}^n$ and spheres are connected.

A simply ordered set L with more than one element is a **linear continuum** if it has the least upper bound property and for $x < y$ there is z with $x < z < y$.

Theorem 3.7 (24.1)

A linear continuum L in the order topology is connected, and so are its intervals and rays.

Proof. It suffices to show convex subsets Y are connected. Suppose A, B separate Y, $a \in A$, $b \in B$, $a < b$. $[a, b] \subset Y$, so $A_0 = A \cap [a, b]$, $B_0 = B \cap [a, b]$ separate $[a, b]$. Let $c = \sup A_0$. If $c \in B_0$, then $c \neq a$, so $c > a$; B_0 is open in $[a, b]$, so some $(d, c] \subset B_0$, and then d is a smaller upper bound for A_0 (since $(d, c]$ misses A_0 and c is the sup): contradiction. If $c \in A_0$, then $c < b$ and some $[c, e) \subset A_0$; choosing z with $c < z < e$ gives $z \in A_0$, contradicting $c = \sup A_0$. ■

Corollary 3.8 (24.2)

$\mathbb{R}$ is connected, and so are its intervals and rays.

Theorem 3.9 (24.3, intermediate value theorem)

Let $f : X \to Y$ be continuous, X connected, Y ordered in the order topology. If $a, b \in X$ and r lies between $f(a)$ and $f(b)$, then $f(c) = r$ for some $c \in X$.

Proof. Otherwise $f(X) \cap (-\infty, r)$ and $f(X) \cap (r, +\infty)$ separate the connected set $f(X)$. ■

Other linear continua: $I \times I$ in the dictionary order (the ordered square), and the **long line** $L = S_\Omega \times [0, 1)$. A **path** from x to y is a continuous $f : [a, b] \to X$ with $f(a) = x$, $f(b) = y$; X is **path connected** if every two points are joined by a path. Path connected spaces are connected (Theorems 3.3, 3.5, 3.8).

Example 3.1

(a) The unit ball B^n and punctured space $\mathbb{R}^n - \{0\}$ $(n > 1)$ are path connected, hence so is S^{n-1} (its image under $x \mapsto x/\|x\|$).
(b) The ordered square is connected but not path connected: a path from the least to the greatest element would be surjective, and the uncountably many disjoint open intervals $U_x = (x \times 0, x \times 1)$ would have disjoint open preimages each containing a rational.
(c) The **topologist's sine curve** $\overline{S}$, the closure in $\mathbb{R}^2$ of $S = \{(x, \sin(1/x)) : 0 < x \leq 1\}$, is

connected (Theorem 3.4) but not path connected: if $f : [a, c] \to \overline{S}$ were a path from 0×0 into S, let $b = \sup\{t : f(t) \in 0 \times [-1, 1]\}$; on $(b, c]$, $f(t) \in S$, and choosing $t_n \downarrow b$ with $\sin(1/x(t_n)) = (-1)^n$ contradicts the continuity of the second coordinate at b.

3.3 Components and local connectedness

Motivation. A disconnected space breaks into maximal connected pieces, its components. Counting components is the simplest topological invariant, used for instance to show that $\mathbb{R}$ and $\mathbb{R}^2$ are not homeomorphic.

Define $x \sim y$ if some connected subspace contains both; the classes are the **components** of X.

Theorem 3.10 (25.1)

The components of X are connected disjoint subspaces whose union is X, and each nonempty connected subspace meets only one of them.

Proof. $\sim$ is an equivalence relation (transitivity by Theorem 3.3). A connected A meeting components C_1, C_2 in x_1, x_2 gives $x_1 \sim x_2$. For connectedness of a component C, fix $x_0 \in C$; each $x \in C$ lies in a connected $A_x \ni x_0$, contained in C by the previous sentence, and $C = \bigcup A_x$ is connected by Theorem 3.3. ■

Similarly, define $x \sim y$ if a path joins them; the classes are the **path components**.

Theorem 3.11 (25.2)

The path components of X are path-connected disjoint subspaces with union X, and each nonempty path-connected subspace meets only one of them.

Proof. Transitivity: concatenate paths $f : [a, b] \to X$ and $g : [b, c] \to X$ via the pasting lemma. ■

Components are closed (the closure of a connected set is connected); path components need not be (the topologist's sine curve has two path components, S and $0 \times [-1, 1]$, the first not closed). X is **locally connected at** x if every neighbourhood of x contains a connected neighbourhood of x; similarly **locally path connected**. Intervals in $\mathbb{R}$ are both; $\mathbb{Q}$ is neither; the topologist's sine curve is connected but not locally connected.

Theorem 3.12 (25.3)

X is locally connected iff for every open U each component of U is open in X.

Proof. ($\Rightarrow$) If C is a component of U and $x \in C$, a connected neighbourhood $V \subset U$ of x lies in C.

($\Leftarrow$) For a neighbourhood U of x, the component of U containing x is a connected neighbourhood inside U. ■

Theorem 3.13 (25.4)

X is locally path connected iff for every open U each path component of U is open in X.

Theorem 3.14 (25.5)

Each path component of X lies in a component. If X is locally path connected, components and path components coincide.

Proof. The first claim since path connected sets are connected. Let C be a component and $P \subset C$ a path component. The path components of X are open (Theorem 3.13 with $U = X$); Q, the union of the other path components meeting C, is open, so P, Q would separate C unless $Q = \emptyset$. Hence $P = C$. ■

3.4 Compact spaces

Motivation. On $[a, b]$ continuous functions are bounded and attain their extrema, and they are uniformly continuous. The property responsible is compactness: every open cover has a finite subcover. It lets us pass from local information to global information, and it is preserved by continuous maps and products.

A collection of subsets **covers** X if its union is X; it is an **open covering** if its members are open. X is **compact** if every open covering contains a finite subcollection covering X. Examples: $\mathbb{R}$ is not compact; $\{0\} \cup \{1/n\}$ is compact; finite spaces and spaces with the finite complement topology are compact.

Lemma 3.15 (26.1)

A subspace $Y \subset X$ is compact iff every covering of Y by sets open in X contains a finite subcollection covering Y.

Proof. Open sets of Y are intersections $Y \cap A$ with A open in X; translate coverings in both directions. ■

Theorem 3.16 (26.2)

Every closed subspace of a compact space is compact.

Proof. Given a covering $\mathcal{A}$ of a closed Y by sets open in X, $\mathcal{A} \cup \{X - Y\}$ covers X; a finite subcover, minus $X - Y$, covers Y. ■

Theorem 3.17 (26.3)

Every compact subspace of a Hausdorff space is closed.

Lemma 3.18 (26.4)

If Y is a compact subspace of a Hausdorff space X and $x_0 \notin Y$, there are disjoint open $U \ni x_0$ and $V \supset Y$.

Proof. For each $y \in Y$ choose disjoint neighbourhoods $U_y \ni x_0$, $V_y \ni y$. Finitely many $V_{y_1}, \ldots, V_{y_n}$ cover Y; take $V = \bigcup V_{y_i}$ and $U = \bigcap U_{y_i}$. This proves Theorem 3.17 too: $X - Y$ is open. ■

Theorem 3.19 (26.5)

The image of a compact space under a continuous map is compact.

Proof. If $\mathcal{A}$ covers $f(X)$ by open sets of Y, $\{f^{-1}(A)\}$ covers X; finitely many suffice. ■

Theorem 3.20 (26.6)

A continuous bijection $f : X \to Y$ from a compact space to a Hausdorff space is a homeomorphism.

Proof. For closed $A \subset X$: A is compact (Theorem 3.16), $f(A)$ is compact (3.19), hence closed (3.17). So f^{-1} is continuous. ■

Lemma 3.21 (26.8, the tube lemma)

If Y is compact and N is an open set of $X \times Y$ containing the slice $x_0 \times Y$, then N contains a tube $W \times Y$ with W a neighbourhood of x_0.

Proof. Cover $x_0 \times Y$ by basis elements $U \times V \subset N$ meeting it; it is compact (homeomorphic to Y), so finitely many $U_i \times V_i$ cover it. Let $W = \bigcap U_i$. Every $x \times y$ with $x \in W$ lies in some $U_i \times V_i$ (choose i with $y \in V_i$), so $W \times Y \subset N$. ■

Theorem 3.22 (26.7)

A product of finitely many compact spaces is compact.

Proof. Let $\mathcal{A}$ be an open cover of $X \times Y$. For each x_0, finitely many members cover $x_0 \times Y$; their union N contains a tube $W_{x_0} \times Y$. Finitely many W_{x_i} cover X, and the corresponding finitely many members of $\mathcal{A}$ cover $X \times Y$. ■

A collection $\mathcal{C}$ has the **finite intersection property** if every finite subcollection has nonempty intersection.

Theorem 3.23 (26.9)

X is compact iff every collection $\mathcal{C}$ of closed sets with the finite intersection property has $\bigcap_{C \in \mathcal{C}} C \neq \emptyset$.

Proof. Pass to complements: $\{X - C\}$ is an open cover iff $\bigcap C = \emptyset$, and a finite subcollection covers iff the corresponding finite intersection is empty. ■

In particular nested sequences $C_1 \supset C_2 \supset \cdots$ of nonempty closed sets in a compact space have nonempty intersection.

3.5 Compact subspaces of the real line

Motivation. We now prove that $[a, b]$ really is compact, and hence that closed bounded sets in $\mathbb{R}^n$ are compact. The extreme value theorem, the Lebesgue number lemma and uniform continuity all follow.

Theorem 3.24 (27.1)

If X is simply ordered with the least upper bound property, every closed interval $[a, b]$ in X is compact (order topology).

Proof. Let $\mathcal{A}$ be a covering of $[a, b]$ by sets open in $[a, b]$.

Step 1. for $x \in [a, b)$ there is $y > x$ such that $[x, y]$ is covered by at most two members of $\mathcal{A}$ (if x has an immediate successor y, use the members containing x, y; otherwise take $A \ni x$ containing $[x, c)$ and any $y \in (x, c)$).

Step 2. let C be the set of $y > a$ in $[a, b]$ such that $[a, y]$ is finitely covered; $C \neq \emptyset$ by Step 1. Let $c = \sup C$.

Step 3. $c \in C$: choose $A \ni c$ containing some $(d, c]$; if $c \notin C$, some $z \in C \cap (d, c)$, and $[a, z]$ plus A covers $[a, c]$.

Step 4. $c = b$: otherwise Step 1 extends past c. ■

Corollary 3.25 (27.2)

Every closed interval in $\mathbb{R}$ is compact.

Theorem 3.26 (27.3)

$A \subset \mathbb{R}^n$ is compact iff it is closed and bounded (in the euclidean or square metric).

Proof. If A is compact, it is closed (Theorem 3.17) and covered by finitely many open balls $B_\rho(0, m)$, hence bounded. Conversely, a closed bounded A lies in a cube $[-P, P]^n$, compact by Theorems 3.22 and 3.25; apply Theorem 3.16. ■

Theorem 3.27 (27.4, extreme value theorem)

If $f : X \to Y$ is continuous, X compact and Y ordered, there are c, d with $f(c) \leq f(x) \leq f(d)$ for all x.

Proof. $A = f(X)$ is compact. If A had no largest element, the rays $(-\infty, a)$, $a \in A$, would cover A; a finite subcover $(-\infty, a_i)$ misses the largest a_i. So A has a largest element $f(d)$; similarly a smallest. ■

For $A \subset X$ in a metric space, $d(x, A) = \inf_{a \in A} d(x, a)$ is continuous in x (indeed $|d(x, A) - d(y, A)| \leq d(x, y)$).

Lemma 3.28 (27.5, Lebesgue number lemma)

Let $\mathcal{A}$ be an open covering of a compact metric space X. There is $\delta > 0$ such that every subset of diameter less than δ lies in some member of $\mathcal{A}$.

Proof. If $X \in \mathcal{A}$ take any δ. Otherwise choose a finite subcover $A_1, \ldots, A_n$, let $C_i = X - A_i$ and $f(x) = \frac{1}{n} \sum d(x, C_i)$. $f > 0$ (each x lies in some A_i, so $d(x, C_i) > 0$) and continuous, so $f \geq \delta > 0$ (Theorem 3.27). If $\operatorname{diam} B < \delta$ and $x_0 \in B$, then $B \subset B(x_0, \delta)$; choosing m with $d(x_0, C_m)$ maximal gives $d(x_0, C_m) \geq f(x_0) \geq \delta$, so $B(x_0, \delta) \subset A_m$. ■

Theorem 3.29 (27.6, uniform continuity theorem)

A continuous map from a compact metric space to a metric space is uniformly continuous.

Proof. Given ϵ, cover X by the open sets $f^{-1}(B(y, \epsilon/2))$ and let δ be a Lebesgue number. If $d(x_1, x_2) < \delta$, $\{x_1, x_2\}$ lies in one of these sets, so $d(f(x_1), f(x_2)) < \epsilon$. ■

A point x is **isolated** if $\{x\}$ is open.

Theorem 3.30 (27.7)

A nonempty compact Hausdorff space without isolated points is uncountable.

Proof.

Step 1. for any nonempty open U and point x, there is a nonempty open $V \subset U$ with $x \notin \overline{V}$ (choose $y \in U$, $y \neq x$, which exists since $U \neq \{x\}$; separate x and y by disjoint $W_1 \ni x$, $W_2 \ni y$; $V = W_2 \cap U$).

Step 2. given $f : \mathbb{Z}_+ \to X$, apply Step 1 recursively to get nonempty open $V_1 \supset V_2 \supset \cdots$ with $f(n) \notin \overline{V_n}$ and $\overline{V_{n+1}} \subset V_n$. The nested closed sets $\overline{V_n}$ have a common point x (Theorem 3.23), and $x \neq f(n)$ for all n. ■

Corollary 3.31 (27.8)

Every closed interval in $\mathbb{R}$ is uncountable.

3.6 Limit point compactness

Motivation. In analysis compactness is often defined with sequences (Bolzano–Weierstrass). We compare the cover, limit-point and sequential versions and show that they agree for metric spaces.

X is **limit point compact** if every infinite subset has a limit point, and **sequentially compact** if every sequence has a convergent subsequence.

Theorem 3.32 (28.1)

Compactness implies limit point compactness, but not conversely.

Proof. If A has no limit point, A is closed and hence compact, and each $a \in A$ has a neighbourhood U_a with $U_a \cap A = \{a\}$. The U_a cover the compact A, finitely many suffice, and each contains one point of A: A is finite. Counterexample to the converse: $X = \mathbb{Z}_+ \times Y$ with Y a two-point indiscrete space (every point has a limit point), or S_Ω below. ■

Theorem 3.33 (28.2)

For a metrizable space X the following are equivalent:
(1) compact;
(2) limit point compact;
(3) sequentially compact.

Proof. (1)⇒(2) is Theorem 3.32. (2)⇒(3): if the range of (x_n) is finite, a constant subsequence converges; otherwise it has a limit point x, and choosing $x_{n_k} \in B(x, 1/k)$ with n_k increasing (possible since each ball contains infinitely many points of the range, Theorem 2.19) gives a convergent

subsequence. (3)⇒(1): first, the Lebesgue number lemma holds: if not, there are C_n of diameter $< 1/n$ in no member of $\mathcal{A}$; pick $x_n \in C_n$ and a convergent subsequence $x_{n_i} \to a$; $a \in A \in \mathcal{A}$ with $B(a,\epsilon) \subset A$, and for large i, $C_{n_i} \subset B(a,\epsilon)$, a contradiction. Second, X is **totally bounded**: if some ϵ admits no finite cover by ϵ-balls, choose $x_{n+1} \notin \bigcup_{i\le n} B(x_i,\epsilon)$; the sequence has no convergent subsequence since all distances are $\ge \epsilon$. Finally, with δ a Lebesgue number, cover X by finitely many $\delta/3$-balls; each lies in a member of $\mathcal{A}$. ■

Example 3.2

S_Ω is limit point compact but not compact: an infinite subset contains a countably infinite subset B, which has an upper bound b (Theorem 1.27), so $B \subset [a_0, b]$, which is compact (Theorem 3.24); hence B has a limit point. S_Ω is not compact since the sections S_α (or the open intervals $[a_0, \alpha)$) cover it without a finite subcover. It is also sequentially compact.

3.7 Local compactness

Motivation. $\mathbb{R}^n$ is not compact, but it is compact near every point, and adding one point at infinity makes it compact ($\mathbb{R}^2 \cup \{\infty\}$ is the sphere). Local compactness is exactly the condition for this one-point compactification to be Hausdorff.

X is **locally compact at** x if some compact C contains a neighbourhood of x, and locally compact if this holds at every point. $\mathbb{R}$, $\mathbb{R}^n$, every simply ordered set with the least upper bound property are locally compact; $\mathbb{Q}$ and $\mathbb{R}^\omega$ are not.

Theorem 3.34 (29.1)

X is locally compact Hausdorff iff there is a space Y with:
(1) X is a subspace of Y;
(2) $Y - X$ is a single point;
(3) Y is compact Hausdorff. Two such Y are homeomorphic by a homeomorphism fixing X.

Proof. Uniqueness: map $Y \to Y'$ by the identity on X and $p \mapsto q$; open sets of Y containing p are complements of compact subsets of X ($Y - U$ is closed in Y, hence compact, and lies in X), and the same holds in Y'.

Existence: let $Y = X \cup \{\infty\}$ with open sets the open sets of X and the sets $Y - C$ for compact $C \subset X$. This is a topology (uses that compact sets in the Hausdorff X are closed); X is a subspace; Y is compact (a cover contains some $Y - C$, and C needs finitely many more); Y is Hausdorff (separate x from ∞ by $\operatorname{Int} C \ni x$ and $Y - C$, where C is a compact set containing a neighbourhood of x). Conversely, a subspace of a compact Hausdorff space with one point removed is Hausdorff and locally compact (separate x from the removed point by disjoint open $U \ni x$, $V \ni \infty$; $C = Y - V$ is compact and contains U). ■

Y is the **one-point compactification**; e.g. $\mathbb{R} \cup \{\infty\} \cong S^1$ and $\mathbb{R}^n \cup \{\infty\} \cong S^n$ (stereographic projection).

Theorem 3.35 (29.2)

A Hausdorff space X is locally compact iff for every x and neighbourhood U of x there is a neighbourhood V of x with $\overline{V}$ compact and $\overline{V} \subset U$.

Proof. ($\Leftarrow$) Take $C = \overline{V}$.

($\Rightarrow$) In the one-point compactification Y, $C = Y - U$ is closed, hence compact; by Lemma 3.18 there are disjoint open $V \ni x$, $W \supset C$. Then $\overline{V}$ (closure in Y) misses W, so $\overline{V} \subset U$; and $\overline{V}$ is compact as a closed subset of Y. ■

Corollary 3.36 (29.3)

A closed or open subspace of a locally compact Hausdorff space is locally compact.

Proof. Closed A: if C is compact containing a neighbourhood U of x, then $C \cap A$ is compact and contains $U \cap A$. Open A: by Theorem 3.35, $x \in V \subset \overline{V} \subset A$ with $\overline{V}$ compact. ■

Corollary 3.37 (29.4)

X is homeomorphic to an open subspace of a compact Hausdorff space iff X is locally compact Hausdorff.

Chapter 4. Countability and Separation Axioms

4.1 The countability axioms

Motivation. Which spaces come from a metric? The answer will be: regular spaces with a countable basis. This chapter builds towards that metrization theorem, starting with the countability axioms, which say that a space can be described by countably many open sets.

X has a **countable basis at** x if some countable collection $\mathcal{B}$ of neighbourhoods of x has every neighbourhood of x containing a member of $\mathcal{B}$; X is **first countable** if this holds at every point. Metric spaces are first countable.

Theorem 4.1 (30.1)

(a) If a sequence of points of A converges to x, then $x \in \overline{A}$; the converse holds if X is first countable.
(b) If $f : X \to Y$ is continuous, it preserves convergent sequences; the converse holds if X is first countable.

Proof. As in Lemma 2.38 and Theorem 2.39: with a countable basis $B_1, B_2, \dots$ at x, replace it by $B_1 \cap \cdots \cap B_n$ (nested) and pick $x_n \in A \cap B_1 \cap \cdots \cap B_n$. ■

X is **second countable** if its topology has a countable basis. $\mathbb{R}^n$ (rational balls) and $\mathbb{R}^\omega$ (products of rational intervals in finitely many coordinates) are second countable; $\mathbb{R}^\omega$ in the uniform topology is first but not second countable (the sets $\{0,1\}^\omega$ give uncountably many points at mutual distance 1).

Theorem 4.2 (30.2)

Subspaces and countable products of first (resp. second) countable spaces are first (resp. second) countable.

Proof. Intersect a countable basis with the subspace. For $\prod_n X_n$ with countable bases $\mathcal{B}_n$, the products $\prod U_n$ with $U_n \in \mathcal{B}_n$ for finitely many n and $U_n = X_n$ otherwise form a countable basis (a countable union of countable sets, Theorem 1.15); similarly at a point. ■

A is **dense** if $\overline{A} = X$.

Theorem 4.3 (30.3)

If X has a countable basis, then (a) every open covering has a countable subcovering (X is **Lindelöf**), and (b) X has a countable dense subset (X is **separable**).

Proof. (a) For each basis element B_n that lies in some member of the cover, choose such a member A_n; the A_n cover X, since each x lies in some $B_n \subset A$ for A in the cover.

(b) Pick $x_n \in B_n$ (nonempty ones); every nonempty open set contains some B_n, hence some x_n. ■

Example 4.1

$\mathbb{R}_\ell$ is first countable, Lindelöf and separable but not second countable (any basis must contain, for each x, a set B_x with $x = \inf B_x$). $\mathbb{R}_\ell^2$ (the Sorgenfrey plane) is not Lindelöf: the antidiagonal $L = \{x \times (-x)\}$ is closed and each point of L has a neighbourhood $[a,b) \times [-a,d)$ meeting L only there; adding $X - L$ gives a cover with no countable subcover. So products of

Lindelöf spaces need not be Lindelöf, and subspaces of Lindelöf spaces need not be Lindelöf. For metrizable spaces, second countable, separable and Lindelöf are equivalent.

4.2 The separation axioms

Motivation. Metric spaces can separate points from closed sets and closed sets from each other by disjoint open sets. The separation axioms (Hausdorff, regular, normal) isolate these properties and measure how far a general space is from being metric.

Assume one-point sets are closed. X is **regular** if for each point x and closed set B not containing x there are disjoint open sets containing x and B; **normal** if each pair of disjoint closed sets has disjoint open neighbourhoods. Normal $\Rightarrow$ regular $\Rightarrow$ Hausdorff.

Lemma 4.4 (31.1)

(a) X is regular iff for each x and neighbourhood U of x there is a neighbourhood V of x with $\overline{V} \subset U$.
(b) X is normal iff for each closed A and open $U \supset A$ there is an open $V \supset A$ with $\overline{V} \subset U$.

Proof. (a) ($\Rightarrow$) Separate x from $B = X - U$ by disjoint $V \ni x$, $W \supset B$; then $\overline{V} \cap B = \emptyset$ since $X - W$ is closed and contains V.

($\Leftarrow$) For $x \notin B$, take $U = X - B$, then V and $X - \overline{V}$ separate.

(b) Same with A for x. ■

Theorem 4.5 (31.2)

(a) Subspaces and products of Hausdorff spaces are Hausdorff.
(b) Subspaces and products of regular spaces are regular.

Proof. (b) Subspace Y: if $x \in Y$ and B closed in Y with $x \notin B$, then $B = \overline{B} \cap Y$ and $x \notin \overline{B}$; separate x from $\overline{B}$ in X and intersect with Y. Product: given $x = (x_\alpha)$ and a basis neighbourhood $\prod U_\alpha$, choose $V_\alpha \ni x_\alpha$ with $\overline{V_\alpha} \subset U_\alpha$ (and $V_\alpha = X_\alpha$ when $U_\alpha = X_\alpha$); then $V = \prod V_\alpha$ has $\overline{V} = \prod \overline{V_\alpha} \subset \prod U_\alpha$ (Theorem 2.30). Apply Lemma 4.4. ■

Example 4.2

(a) $\mathbb{R}_K$ is Hausdorff but not regular: K is closed and $0 \notin K$, but any neighbourhood of 0 has closure meeting every open set containing K.
(b) $\mathbb{R}_\ell$ is normal, but $\mathbb{R}_\ell^2$ is regular (a product of regular spaces) and not normal (Example 4.3). So products of normal spaces need not be normal.

4.3 Normal spaces

Motivation. Normality, separating disjoint closed sets, is the property needed to construct continuous functions, but it is badly behaved under products and subspaces. Here we find the main classes of normal spaces: metric spaces, compact Hausdorff spaces, and regular spaces with a countable basis.

Theorem 4.6 (32.1)

Every regular space with a countable basis is normal.

Proof. Let $\mathcal{B}$ be a countable basis and A, B disjoint closed. Each $a \in A$ has a basis element $U \ni a$ with $\overline{U} \cap B = \emptyset$ (regularity, Lemma 4.4); enumerate those used as $U_1, U_2, \ldots$; they cover A and their closures miss B. Similarly $V_1, V_2, \ldots$ cover B with closures missing A. Put $U_n' = U_n - \bigcup_{i \le n} \overline{V_i}$, $V_n' = V_n - \bigcup_{i \le n} \overline{U_i}$, which are open. $U' = \bigcup U_n' \supset A$ and $V' = \bigcup V_n' \supset B$. If $x \in U_j' \cap V_k'$ with $j \le k$, then $x \in U_j$ but $x \notin \overline{U_j}$ (from V_k'), a contradiction; similarly for $j > k$. ∎

Theorem 4.7 (32.2)

Every metrizable space is normal.

Proof. For disjoint closed A, B choose for each $a \in A$ an ϵ_a with $B(a, \epsilon_a) \cap B = \emptyset$, and for each $b \in B$ an ϵ_b with $B(b, \epsilon_b) \cap A = \emptyset$. Let $U = \bigcup_a B(a, \epsilon_a/2)$, $V = \bigcup_b B(b, \epsilon_b/2)$. If $z \in U \cap V$ with $d(a, z) < \epsilon_a/2$, $d(b, z) < \epsilon_b/2$, then $d(a, b) < (\epsilon_a + \epsilon_b)/2 \le \max(\epsilon_a, \epsilon_b)$, contradicting the choice of radii. ∎

Theorem 4.8 (32.3)

Every compact Hausdorff space is normal.

Proof. Regularity: Lemma 3.18 with the compact closed set B. Normality: for disjoint closed A, B, choose for each $a \in A$ disjoint $U_a \ni a$, $V_a \supset B$; finitely many U_{a_i} cover A; take $U = \bigcup U_{a_i}$, $V = \bigcap V_{a_i}$. ∎

Theorem 4.9 (32.4)

Every well-ordered set is normal in the order topology.

Proof. Every interval $(x, y]$ is open (it equals (x, y^+) with y^+ the successor of y, or a ray if y is largest). Let A, B be disjoint closed, first not containing the smallest element a_0. For $a \in A$ choose $(x_a, a]$ missing B, and for $b \in B$ choose $(y_b, b]$ missing A. $U = \bigcup (x_a, a]$, $V = \bigcup (y_b, b]$ are disjoint: if $z \in (x_a, a] \cap (y_b, b]$ with $a < b$, then $y_b < z \le a < b$, so $a \in (y_b, b]$, which misses A. If $a_0 \in A$, note $\{a_0\}$ is open and closed; apply the argument to $A - \{a_0\}$ and add $\{a_0\}$ to U. ∎

Example 4.3

(a) The Sorgenfrey plane $\mathbb{R}_\ell^2$ is not normal. With L the antidiagonal, every subset $A \subset L$ is closed, as is $L - A$. If each pair A, $L - A$ had disjoint open neighbourhoods $U_A \supset A$, $V_A \supset L - A$, then $A \mapsto U_A \cap D$ (D the rational points) would be injective from $\mathcal{P}(L)$ into $\mathcal{P}(D)$, which has the cardinality of $\mathcal{P}(\mathbb{Z}_+)$; but L has the cardinality of $\mathbb{R}$, and $\mathcal{P}(\mathbb{R})$ is strictly larger (Theorem 1.18).
(b) $S_\Omega \times \overline{S}_\Omega$ is not normal although both factors are (Theorem 4.9): the diagonal $A = \{x \times x\}$ and $B = S_\Omega \times \{\Omega\}$ are disjoint closed sets that cannot be separated. Since $\overline{S}_\Omega \times \overline{S}_\Omega$ is compact Hausdorff, hence normal, normality is not hereditary.
(c) $\mathbb{R}^J$ for uncountable J is not normal.

4.4 The Urysohn lemma

Motivation. In a metric space, $d(x,A)/(d(x,A)+d(x,B))$ is a continuous function that is 0 on A and 1 on B. Urysohn's lemma produces such a function in any normal space, without a metric. It is the first deep theorem of the course and the engine of both the metrization and the extension theorems.

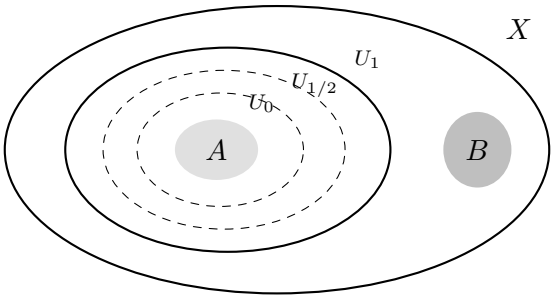


Figure. *Urysohn's construction: open sets U_r, one for each rational $r \in [0,1]$, with $A \subset U_0$, $\overline{U_r} \subset U_s$ for $r<s$, and $U_1 = X - B$. Then $f(x) = \inf\{r : x \in U_r\}$ is continuous, 0 on A and 1 on B.*

Theorem 4.10 (33.1, Urysohn lemma)

Let X be normal and A, B disjoint closed subsets. For any $[a,b] \subset \mathbb{R}$ there is a continuous $f : X \to [a,b]$ with $f = a$ on A and $f = b$ on B.

Proof. It suffices to treat $[0,1]$.

Step 1. let P be the rationals in $[0,1]$, enumerated with $1, 0$ first. Define open U_p $(p \in P)$ such that $p<q \Rightarrow \overline{U_p} \subset U_q$: put $U_1 = X - B$; by Lemma 4.4(b) choose $U_0 \supset A$ with $\overline{U_0} \subset U_1$. If U_p is defined for the first n rationals P_n with the property, let r be the next; $P_n \cup \{r\}$ is finite and ordered, r has an immediate predecessor p and successor q in it, and normality gives U_r with $\overline{U_p} \subset U_r \subset \overline{U_r} \subset U_q$. By induction (recursive definition) all U_p are defined.

Step 2. extend to all rationals p by $U_p = \emptyset$ for $p<0$ and $U_p = X$ for $p>1$; the property persists.

Step 3. let $\mathbb{Q}(x) = \{p : x \in U_p\}$, which contains no $p<0$ and every $p>1$, and $f(x) = \inf \mathbb{Q}(x) \in [0,1]$. On A, $f=0$; on B, $x \notin U_p$ for $p \le 1$, so $f=1$.

Step 4. (i) $x \in \overline{U_r} \Rightarrow f(x) \le r$ (then $x \in U_s$ for all rational $s>r$);

(ii) $x \notin U_r \Rightarrow f(x) \ge r$ (then $x \notin U_s$ for $s<r$).

Step 5. continuity: given x_0 and $(c,d) \ni f(x_0)$, choose rationals $c<p<f(x_0)<q<d$. Then $U = U_q - \overline{U_p}$ is an open neighbourhood of x_0 (by (i), (ii) contrapositives), and for $x \in U$: $x \in U_q \subset \overline{U_q}$ gives $f(x) \le q$; $x \notin \overline{U_p} \supset U_p$ gives $f(x) \ge p$. So $f(U) \subset [p,q] \subset (c,d)$. ∎

A, B are **separated by a continuous function** if there is $f : X \to [0,1]$ with $f(A) = \{0\}$, $f(B) = \{1\}$. A space in which one-point sets are closed and each point x_0 and closed set $A \not\ni x_0$ are so separated is **completely regular**. Normal $\Rightarrow$ completely regular $\Rightarrow$ regular (if $f(x_0) = 0$, $f(A) = 1$, the sets $f^{-1}([0, \frac{1}{2}))$, $f^{-1}((\frac{1}{2}, 1])$ separate).

Theorem 4.11 (33.2)

Subspaces and products of completely regular spaces are completely regular.

Proof. Subspace: restrict the function separating x_0 from $\overline{A}$. Product: given $x = (x_\alpha)$ and a basis neighbourhood $\prod U_\alpha$ with $U_\alpha \ne X_\alpha$ only for $\alpha_1, \dots, \alpha_n$, take $f_i : X_{\alpha_i} \to [0,1]$ with $f_i(x_{\alpha_i}) = 1$ and $f_i = 0$ off U_{α_i}; then $f = \prod_i f_i \circ \pi_{\alpha_i}$ is 1 at x and 0 off $\prod U_\alpha$. ∎

So $\mathbb{R}_\ell^2$ and $S_\Omega \times \overline{S}_\Omega$ are completely regular but not normal.

4.5 The Urysohn metrization theorem

Motivation. Combining Urysohn functions with the countable basis, we embed the space in $\mathbb{R}^\omega$, which is metrizable. This answers the metrization question for second countable spaces: they are metrizable exactly when they are regular.

Theorem 4.12 (34.1, Urysohn metrization theorem)

Every regular space with a countable basis is metrizable.

Proof.

Step 1. there is a countable family $f_n : X \to [0,1]$ such that for each x_0 and neighbourhood U of x_0 some f_n is positive at x_0 and vanishes off U. Indeed X is normal (Theorem 4.6); for each pair (n,m) of indices with $\overline{B_n} \subset B_m$ (B_i the countable basis), Urysohn gives $g_{n,m} = 1$ on $\overline{B_n}$, 0 off B_m. Given $x_0 \in U$, choose B_m with $x_0 \in B_m \subset U$ and, by regularity, B_n with $x_0 \in B_n$, $\overline{B_n} \subset B_m$; $g_{n,m}$ works.

Step 2. by the imbedding theorem below, $F(x) = (f_1(x), f_2(x), \dots)$ imbeds X in $\mathbb{R}^\omega$ (product topology), which is metrizable (Theorem 2.36). (One may also give a direct metric $D(F(x), F(y)) = \sup|f_n(x) - f_n(y)|/n$.) ■

Theorem 4.13 (34.2, imbedding theorem)

Let X be a space in which one-point sets are closed and $\{f_\alpha\}_{\alpha \in J}$ continuous functions $X \to \mathbb{R}$ such that for each x_0 and neighbourhood U of x_0 some f_α is positive at x_0 and zero outside U. Then $F(x) = (f_\alpha(x))_{\alpha \in J}$ is an imbedding of X in $\mathbb{R}^J$. If each f_α maps into $[0,1]$, F imbeds X in $[0,1]^J$.

Proof. F is continuous (Theorem 2.31). Injective: for $x \neq y$, choose α with $f_\alpha(x) > 0$ and $f_\alpha = 0$ off $X - \{y\}$ (open since points are closed), so $f_\alpha(y) = 0$. Open onto $Z = F(X)$: let U be open in X and $z_0 = F(x_0) \in F(U)$. Choose β with $f_\beta(x_0) > 0$, $f_\beta = 0$ off U, and let $W = \pi_\beta^{-1}((0,\infty)) \cap Z$, open in Z and containing z_0. If $z = F(x) \in W$, then $f_\beta(x) > 0$, so $x \in U$ and $z \in F(U)$. Thus $F(U)$ is open in Z. ■

Corollary 4.14 (34.3)

X is completely regular iff it is homeomorphic to a subspace of $[0,1]^J$ for some J.

Proof. $[0,1]^J$ is completely regular (Theorem 4.11), hence so are its subspaces. Conversely, for completely regular X the family of all continuous $f : X \to [0,1]$ satisfies the hypothesis of Theorem 4.13 (separate x_0 from $X - U$). ■

4.6 The Tietze extension theorem

Motivation. Can a continuous function defined on a closed set be extended to the whole space? In normal spaces it always can; the proof builds the extension as a uniform limit of Urysohn functions.

Theorem 4.15 (35.1, Tietze extension theorem)

Let X be normal and $A \subset X$ closed.
(a) Every continuous $f : A \to [a, b]$ extends to a continuous $g : X \to [a, b]$.
(b) Every continuous $f : A \to \mathbb{R}$ extends to a continuous $g : X \to \mathbb{R}$.

Proof.

Step 1. for continuous $f : A \to [-r, r]$ there is continuous $g : X \to \mathbb{R}$ with $|g| \le r/3$ on X and $|g - f| \le 2r/3$ on A: the sets $B = f^{-1}([-r, -r/3])$, $C = f^{-1}([r/3, r])$ are disjoint and closed in X; Urysohn gives $g : X \to [-r/3, r/3]$ with $g = -r/3$ on B and $r/3$ on C, and one checks the three cases.

Step 2. for $f : A \to [-1, 1]$ apply Step 1 repeatedly: g_1 for f with $r = 1$; g_2 for $f - g_1$ with $r = \frac{2}{3}$; in general g_n with $|g_n| \le \frac{1}{3}(\frac{2}{3})^{n-1}$ and $|f - \sum_{i \le n} g_i| \le (\frac{2}{3})^n$ on A. $g = \sum g_n$ converges uniformly (comparison with a geometric series), so g is continuous (Theorem 2.42), $|g| \le 1$, and $g = f$ on A.

Step 3. for (b), regard f as a map into $(-1, 1) \cong \mathbb{R}$. Extend it by Step 2 to $g : X \to [-1, 1]$. The set $D = g^{-1}(\{-1, 1\})$ is closed and disjoint from A; Urysohn gives $\phi : X \to [0, 1]$ with $\phi = 0$ on D, $\phi = 1$ on A. Then $h = \phi g$ extends f and maps into $(-1, 1)$. ■

Conversely, Tietze's theorem implies the Urysohn lemma (extend the function equal to 0 on A and 1 on B from the closed set $A \cup B$).

4.7 Imbeddings of manifolds

Motivation. Abstract manifolds are defined without reference to any ambient space. Partitions of unity let us embed every compact manifold in some $\mathbb{R}^N$, so abstract manifolds are no more general than submanifolds of Euclidean space.

An **m-manifold** is a Hausdorff space with a countable basis in which each point has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^m$. For $\phi : X \to \mathbb{R}$, the **support** is $\overline{\{\phi \neq 0\}}$. A **partition of unity dominated by** an indexed open covering $\{U_i\}$ is a family of continuous $\phi_i : X \to [0, 1]$ with $\operatorname{supp} \phi_i \subset U_i$ and $\sum \phi_i = 1$.

Theorem 4.16 (36.1, existence of finite partitions of unity)

Every finite open covering $\{U_1, \dots, U_n\}$ of a normal space has a dominated partition of unity.

Proof.

Step 1 (shrinking). there are open V_i covering X with $\overline{V_i} \subset U_i$. Inductively, $A = X - (V_1 \cup \cdots \cup V_{k-1} \cup U_{k+1} \cup \cdots \cup U_n)$ is closed and contained in U_k; choose $V_k \supset A$ with $\overline{V_k} \subset U_k$; the sets $V_1, \dots, V_k, U_{k+1}, \dots, U_n$ still cover.

Step 2. shrink again to W_i with $\overline{W_i} \subset V_i$, and take Urysohn $\psi_i = 1$ on $\overline{W_i}$, 0 off V_i; $\operatorname{supp} \psi_i \subset \overline{V_i} \subset U_i$. $\Psi = \sum \psi_i > 0$ everywhere (the W_i cover), and $\phi_i = \psi_i / \Psi$. ■

Theorem 4.17 (36.2)

Every compact m-manifold imbeds in $\mathbb{R}^N$ for some N.

Proof. Cover X by finitely many open U_i with imbeddings $g_i : U_i \to \mathbb{R}^m$, and let ϕ_i be a dominated partition of unity (X is compact Hausdorff, hence normal). Define $h_i = \phi_i g_i$ on U_i and 0 off $\operatorname{supp} \phi_i$ (continuous by the pasting lemma for open sets), and $F = (\phi_1, \dots, \phi_n, h_1, \dots, h_n) : X \to \mathbb{R}^{n(m+1)}$. F is continuous; injective: if $F(x) = F(y)$, choose i with $\phi_i(x) > 0$; then $\phi_i(y) = \phi_i(x) > 0$, so

$x, y \in U_i$, and $\phi_i(x)g_i(x) = \phi_i(y)g_i(y)$ gives $g_i(x) = g_i(y)$, so $x = y$. A continuous injection of a compact space into a Hausdorff space is an imbedding (Theorem 3.20). ■

Chapter 5. The Tychonoff Theorem

5.1 The Tychonoff theorem

Motivation. Products of compact spaces are compact, even infinite products. This is the key fact behind many existence results in analysis (for example compactness of the unit ball in the weak* topology). The proof requires Zorn's lemma, and in fact Tychonoff's theorem is equivalent to the axiom of choice.

We prove compactness via the finite intersection property (Theorem 3.23): a space is compact iff every collection $\mathcal{A}$ of subsets with the finite intersection property (FIP) satisfies $\bigcap_{A\in\mathcal{A}} \overline{A} \neq \emptyset$.

Lemma 5.1 (37.1)

Let $\mathcal{A}$ be a collection of subsets of X with the FIP. Then there is a collection $\mathcal{D} \supset \mathcal{A}$ with the FIP such that no collection properly containing $\mathcal{D}$ has the FIP ($\mathcal{D}$ is *maximal* with respect to the FIP).

Proof. Let $\mathbb{A}$ be the set of all collections $\mathcal{B} \supset \mathcal{A}$ of subsets of X with the FIP, strictly partially ordered by proper inclusion. If $\mathbb{B} \subset \mathbb{A}$ is a chain, $\mathcal{C} = \bigcup_{\mathcal{B}\in\mathbb{B}} \mathcal{B}$ has the FIP: finitely many $C_1, \ldots, C_n \in \mathcal{C}$ lie in $\mathcal{B}_1, \ldots, \mathcal{B}_n$, all contained in the largest $\mathcal{B}_k$ (chain), which has the FIP. So $\mathcal{C}$ is an upper bound, and Zorn's lemma (Theorem 1.29) gives a maximal element. ■

Lemma 5.2 (37.2)

Let $\mathcal{D}$ be maximal with respect to the FIP.
(a) Every finite intersection of elements of $\mathcal{D}$ is in $\mathcal{D}$.
(b) If $A \subset X$ intersects every element of $\mathcal{D}$, then $A \in \mathcal{D}$.

Proof. (a) Let B be a finite intersection of elements of $\mathcal{D}$; $\mathcal{E} = \mathcal{D} \cup \{B\}$ has the FIP (a finite intersection of members of $\mathcal{E}$ is a finite intersection of members of $\mathcal{D}$), so $\mathcal{E} = \mathcal{D}$ by maximality.

(b) $\mathcal{E} = \mathcal{D} \cup \{A\}$: a finite intersection $A \cap D_1 \cap \cdots \cap D_n$ is nonempty since $D_1 \cap \cdots \cap D_n \in \mathcal{D}$ by (a) and A meets it. So $A \in \mathcal{D}$. ■

Theorem 5.3 (37.3, Tychonoff theorem)

An arbitrary product of compact spaces is compact in the product topology.

Proof. Let $X = \prod_{\alpha\in J} X_\alpha$ and $\mathcal{A}$ a collection of subsets of X with the FIP; extend it to a maximal $\mathcal{D}$ (Lemma 5.1). It suffices to find $x \in \bigcap_{D\in\mathcal{D}} \overline{D}$. For each α, $\{\pi_\alpha(D) : D \in \mathcal{D}\}$ has the FIP in the compact X_α, so choose $x_\alpha \in \bigcap_D \overline{\pi_\alpha(D)}$, and let $x = (x_\alpha)$.

Step 1. if U_β is a neighbourhood of x_β, then $\pi_\beta^{-1}(U_\beta)$ meets every $D \in \mathcal{D}$ (since U_β meets $\pi_\beta(D)$, as $x_\beta \in \overline{\pi_\beta(D)}$), so $\pi_\beta^{-1}(U_\beta) \in \mathcal{D}$ by Lemma 5.2(b).

Step 2. every basis element containing x is a finite intersection of such subbasis elements, hence lies in $\mathcal{D}$ (Lemma 5.2(a)), hence meets every element of $\mathcal{D}$ (FIP). So $x \in \overline{D}$ for all $D \in \mathcal{D}$. ■

The theorem fails for the box topology ($[0,1]^\omega$ in the box topology is not compact), and it is equivalent to the axiom of choice.

Example 5.1

$[0,1]^\omega$ is a compact metrizable space; every regular second countable space imbeds in it (proof of Theorem 4.12). $[0,1]^J$ for uncountable J is compact Hausdorff but not metrizable: it is not even first countable (a countable family of basis neighbourhoods of a point restricts only countably many coordinates).

5.2 The Stone–Čech compactification

Motivation. Every completely regular space can be compactified, and there is a largest compactification βX in which every bounded continuous function on X extends. It is constructed by embedding X into a product of intervals and taking the closure, so Tychonoff's theorem supplies compactness.

A **compactification** of X is a compact Hausdorff space Y containing X as a subspace with $\overline{X} = Y$; two are **equivalent** if there is a homeomorphism between them that is the identity on X. If X has a compactification, it is completely regular (a subspace of a compact Hausdorff, hence normal, space).

Lemma 5.4 (38.1)

If $h : X \to Z$ is an imbedding in a compact Hausdorff space Z, there is a compactification Y of X and an imbedding $H : Y \to Z$ equal to h on X; Y is unique up to equivalence (Y is the compactification **induced** by h).

Proof. Let $X_0 = h(X)$ and $Y_0 = \overline{X_0}$, a compact Hausdorff space. Form $Y = X \sqcup (Y_0 - X_0)$ with the bijection $H : Y \to Y_0$ (h on X, identity elsewhere) and give Y the topology making H a homeomorphism. Then X is a subspace of Y, dense since X_0 is dense in Y_0. ■

Theorem 5.5 (38.2)

Every completely regular space X has a compactification Y such that every bounded continuous $f : X \to \mathbb{R}$ extends uniquely to a continuous $Y \to \mathbb{R}$.

Proof. Let $\{f_\alpha\}_{\alpha \in J}$ be all bounded continuous real functions on X, with $f_\alpha(X) \subset I_\alpha = [\inf f_\alpha, \sup f_\alpha]$. $h = (f_\alpha) : X \to Z = \prod I_\alpha$ is an imbedding (Theorem 4.13, using complete regularity), and Z is compact Hausdorff (Tychonoff). Let Y be the induced compactification with $H : Y \to Z$. For $f = f_\beta$, $\pi_\beta \circ H$ is a continuous extension. Uniqueness follows from the next lemma. ■

Lemma 5.6 (38.3)

If $A \subset X$ and $f : A \to Z$ is continuous with Z Hausdorff, there is at most one continuous extension $g : \overline{A} \to Z$.

Proof. If $g \neq g'$ at x, separate $g(x), g'(x)$ by disjoint U, U'; the neighbourhood $g^{-1}(U) \cap g'^{-1}(U')$ of x contains a point $y \in A$, but $g(y) = g'(y) = f(y)$ lies in $U \cap U'$. ■

Theorem 5.7 (38.4)

If Y is a compactification of the completely regular X with the extension property of Theorem 5.5, every continuous map f of X into a compact Hausdorff space C extends uniquely to

a continuous $g : Y \to C$.

Proof. C is completely regular, so imbeds in some $[0,1]^J$ (Corollary 4.14); regard $C \subset [0,1]^J$. Each coordinate $\pi_\alpha \circ f$ extends to $g_\alpha : Y \to [0,1]$; $g = (g_\alpha) : Y \to [0,1]^J$ is continuous. Since $g(X) = f(X) \subset C$ and X is dense, $g(Y) \subset \overline{g(X)} \subset C$ (C is closed).

Uniqueness: Lemma 5.6. ■

Theorem 5.8 (38.5)

Two compactifications Y_1, Y_2 of X with the extension property of Theorem 5.5 are equivalent.

Proof. Extend the inclusion $j_2 : X \to Y_2$ to $f_2 : Y_1 \to Y_2$ and j_1 to $f_1 : Y_2 \to Y_1$ (Theorem 5.7). $f_1 \circ f_2 : Y_1 \to Y_1$ extends the identity of X, as does id_{Y_1}; by Lemma 5.6, $f_1 \circ f_2 = \mathrm{id}$. Similarly $f_2 \circ f_1 = \mathrm{id}$. ■

This compactification is the **Stone–Čech compactification** $\beta(X)$. It is characterized by the universal property of Theorem 5.7.

Example 5.2

(a) The one-point compactification of $(0,1)$ is the circle and $[0,1]$ is a two-point compactification; neither is $\beta(0,1)$, since $\sin(1/x)$ extends to $\beta(0,1)$.
(b) $\beta(\mathbb{Z}_+)$ is not sequentially compact: $\mathbb{Z}_+$ has no convergent subsequence in $\beta\mathbb{Z}_+$ (separate even and odd subsequences by extending functions on $\mathbb{Z}_+$).
(c) $\beta(S_\Omega) = \overline{S}_\Omega$, because every continuous real function on S_Ω is eventually constant.

Chapter 6. Metrization Theorems and Paracompactness

6.1 Local finiteness

Motivation. Urysohn metrization needs a countable basis; to go further we need a substitute. A locally finite family of sets, one where each point meets only finitely many of them, is the right weakening, since unions and continuous functions defined piece by piece behave well for such families.

A collection $\mathcal{A}$ of subsets of X is **locally finite** if every point has a neighbourhood meeting only finitely many elements of $\mathcal{A}$; it is **countably locally finite** (σ-locally finite) if it is a countable union of locally finite collections. A collection $\mathcal{B}$ **refines** $\mathcal{A}$ (is a **refinement** of $\mathcal{A}$) if each $B \in \mathcal{B}$ lies in some $A \in \mathcal{A}$; it is an open (closed) refinement if its elements are open (closed).

Lemma 6.1 (39.1)

Let $\mathcal{A}$ be locally finite.
(a) Every subcollection of $\mathcal{A}$ is locally finite.
(b) $\{\overline{A} : A \in \mathcal{A}\}$ is locally finite.
(c) $\overline{\bigcup_{A\in\mathcal{A}} A} = \bigcup_{A\in\mathcal{A}} \overline{A}$.

Proof. (a) is clear.

(b) An open set U meeting $\overline{A}$ meets A; so a neighbourhood meeting finitely many A meets only the corresponding $\overline{A}$.

(c) The inclusion $\supset$ always holds. Let $x \in \overline{\bigcup A}$ and let U be a neighbourhood of x meeting only $A_1, \dots, A_k$. If $x \notin \overline{A_1} \cup \cdots \cup \overline{A_k}$, then $U - (\overline{A_1} \cup \cdots \cup \overline{A_k})$ is a neighbourhood of x missing every element of $\mathcal{A}$, a contradiction. ■

Without local finiteness (c) fails: the one-point sets $\{1/n\}$ have union with closure containing 0.

Lemma 6.2 (39.2)

If X is metrizable, every open covering $\mathcal{A}$ of X has a countably locally finite open refinement $\mathcal{E}$ that covers X.

Proof. Fix a metric d and a well-ordering $<$ of $\mathcal{A}$ (Theorem 1.25). For $n \in \mathbb{Z}_+$ and $U \in \mathcal{A}$ let

$$S_n(U) = \{x : B(x, 1/n) \subset U\}, \qquad T_n(U) = S_n(U) - \bigcup_{V<U} V, \qquad E_n(U) = \bigcup_{x\in T_n(U)} B(x, 1/3n).$$

If $V < U$, $x \in T_n(V)$, $y \in T_n(U)$, then $B(x, 1/n) \subset V$ and $y \notin V$, so $d(x, y) \geq 1/n$. Hence distinct sets $E_n(U), E_n(V)$ are at distance at least $1/3n$, a ball of radius $1/6n$ meets at most one of them, and $\mathcal{E}_n = \{E_n(U)\}_U$ is locally finite. Also $E_n(U) \subset U$, since $B(x, 1/3n) \subset B(x, 1/n) \subset U$ for $x \in T_n(U)$. Finally, given x, let U be the first element of $\mathcal{A}$ containing x and choose n with $B(x, 1/n) \subset U$: then $x \in S_n(U)$ and x lies in no $V < U$, so $x \in T_n(U) \subset E_n(U)$. Take $\mathcal{E} = \bigcup_n \mathcal{E}_n$. ■

6.2 The Nagata–Smirnov metrization theorem

Motivation. This gives the complete answer to the metrization problem: a space is metrizable if and only if it is regular with a countably locally finite basis. The proof follows the Urysohn embedding, with a countable basis replaced by a countably locally finite one.

A set is a G_δ **set** if it is a countable intersection of open sets. In a metric space every closed set A is G_δ: $A = \bigcap_n \{x : d(x, A) < 1/n\}$.

Lemma 6.3 (40.1)

Let X be regular with a countably locally finite basis $\mathcal{B}$. Then X is normal and every closed set is a G_δ set.

Proof.

Step 1. for each open W there are open $U_1, U_2, \ldots$ with $W = \bigcup U_n = \bigcup \overline{U_n}$. Write $\mathcal{B} = \bigcup \mathcal{B}_n$ with $\mathcal{B}_n$ locally finite, let $\mathcal{C}_n = \{B \in \mathcal{B}_n : \overline{B} \subset W\}$ (locally finite) and $U_n = \bigcup_{B \in \mathcal{C}_n} B$. By Lemma 6.1(c), $\overline{U_n} = \bigcup_{\mathcal{C}_n} \overline{B} \subset W$. If $x \in W$, regularity gives a basis element $B \ni x$ with $\overline{B} \subset W$; $B \in \mathcal{C}_n$ for some n, so $x \in U_n$.

Step 2. if C is closed, apply Step 1 to $W = X - C$: $C = \bigcap_n (X - \overline{U_n})$, a G_δ.

Step 3. for disjoint closed C, D, Step 1 with $W = X - D$ gives open U_n covering C with closures missing D, and with $W = X - C$ open V_n covering D with closures missing C. Now finish exactly as in the proof of Theorem 4.6. ■

Lemma 6.4 (40.2)

If X is normal and A is a closed G_δ set, there is a continuous $f : X \to [0, 1]$ with $f = 0$ on A and $f > 0$ off A.

Proof. Write $A = \bigcap U_n$ with U_n open. Urysohn gives $f_n : X \to [0, 1]$ equal to 0 on A and 1 on $X - U_n$. Let $f = \sum_n f_n/2^n$, continuous by uniform convergence (Theorem 2.42). It vanishes on A; if $x \notin A$, then $x \notin U_n$ for some n, so $f(x) \geq 2^{-n} > 0$. ■

Theorem 6.5 (40.3, Nagata–Smirnov metrization theorem)

X is metrizable iff X is regular and has a countably locally finite basis.

Proof. ($\Leftarrow$)

Step 1. let $\mathcal{B} = \bigcup \mathcal{B}_n$ with $\mathcal{B}_n$ locally finite. X is normal and $X - B$ is a closed G_δ for each $B \in \mathcal{B}$ (Lemma 6.3), so Lemma 6.4 gives $f_{n,B} : X \to [0, 1/n]$ positive on B and zero off B. For $x_0 \in U$ choose $B \in \mathcal{B}_n$ with $x_0 \in B \subset U$; then $f_{n,B}(x_0) > 0$ and $f_{n,B} = 0$ off U.

Step 2. let $J = \{(n, B) : B \in \mathcal{B}_n\}$ and $F(x) = (f_{n,B}(x)) \in [0, 1]^J$; give $[0, 1]^J$ the uniform metric $\bar{\rho}$. By the proof of Theorem 4.13, F is injective and maps open sets of X to sets open in $F(X)$ for the product topology, hence also for the finer uniform topology (Theorem 2.35).

Step 3 (F is continuous into the uniform metric). let $x_0 \in X$ and $\epsilon > 0$. For each n choose a neighbourhood U_n of x_0 meeting only finitely many elements of $\mathcal{B}_n$; all but finitely many $f_{n,B}$ vanish on U_n, and for the remaining ones continuity gives a smaller neighbourhood W_n on which $|f_{n,B}(x) - f_{n,B}(x_0)| \leq \epsilon/2$ for every $B \in \mathcal{B}_n$. Choose N with $1/N \leq \epsilon/2$ and let $W = W_1 \cap \cdots \cap W_N$. For $x \in W$ the coordinates with $n \leq N$ differ by at most $\epsilon/2$, and those with $n > N$ lie in $[0, 1/n]$, so differ by at most $1/n < \epsilon/2$. Thus $\bar{\rho}(F(x), F(x_0)) \leq \epsilon/2 < \epsilon$, and F imbeds X in a metric space.

($\Rightarrow$) A metrizable space is regular (Theorem 4.7). For each m let $\mathcal{A}_m$ be the covering by balls of radius $1/m$ and $\mathcal{D}_m$ a countably locally finite open refinement covering X (Lemma 6.2); its elements have diameter at most $2/m$. $\mathcal{D} = \bigcup_m \mathcal{D}_m$ is countably locally finite. It is a basis: if $x \in U$ open, choose m with $B(x, 1/m) \subset U$ and $D \in \mathcal{D}_{3m}$ containing x; D has diameter at most $2/3m < 1/m$, so $D \subset B(x, 1/m) \subset U$. ■

6.3 Paracompactness

Motivation. Many constructions in geometry need partitions of unity subordinate to arbitrary open covers. Paracompactness, every open cover has a locally finite refinement, is precisely what makes these possible, and every metric space has it.

X is **paracompact** if every open covering has a locally finite open refinement that covers X. Compact spaces are paracompact, and so is $\mathbb{R}^n$ (Theorem 6.9 below).

Theorem 6.6 (41.1)

Every paracompact Hausdorff space is normal.

Proof. Regularity: let $a \notin B$, B closed. For each $b \in B$ choose open $U_b \ni b$ with $a \notin \overline{U_b}$ (Hausdorff). Cover X by the U_b and $X - B$, take a locally finite open refinement $\mathcal{C}$ covering X, and let $\mathcal{D}$ be the elements of $\mathcal{C}$ meeting B. Each $D \in \mathcal{D}$ is not inside $X - B$, so $D \subset U_b$ for some b and $a \notin \overline{D}$. $V = \bigcup_{\mathcal{D}} D$ is open, contains B, and $\overline{V} = \bigcup \overline{D} \not\ni a$ (Lemma 6.1). Then V and $X - \overline{V}$ separate B and a. Normality: the same argument with a closed set A in place of a, using regularity to choose U_b with $\overline{U_b} \cap A = \emptyset$. ■

Theorem 6.7 (41.2)

Every closed subspace of a paracompact space is paracompact.

Proof. Let Y be closed and $\mathcal{A}$ a covering of Y by sets open in Y; for each $A \in \mathcal{A}$ choose A' open in X with $A' \cap Y = A$. Cover X by the A' together with $X - Y$, and take a locally finite open refinement $\mathcal{B}$ covering X. Then $\{B \cap Y : B \in \mathcal{B}\}$ covers Y, is locally finite, consists of sets open in Y, and refines $\mathcal{A}$ (a B inside $X - Y$ gives the empty set). ■

A paracompact space can have non-paracompact subspaces: $\overline{S}_\Omega$ is compact, but S_Ω is not paracompact (it is not metrizable yet is locally metrizable; see Theorem 6.14). The product $\mathbb{R}^2_\ell$ of paracompact spaces is not paracompact, because it is Hausdorff but not normal.

Lemma 6.8 (41.3, Michael)

Let X be regular. The following are equivalent: every open covering of X has a refinement that is (1) an open covering of X and countably locally finite;
(2) a covering of X and locally finite;
(3) a closed covering of X and locally finite;
(4) an open covering of X and locally finite.

Proof. (4)⇒(1) is trivial.

(1)⇒(2): let $\mathcal{B} = \bigcup \mathcal{B}_n$ be an open refinement as in (1), $V_i = \bigcup_{U \in \mathcal{B}_i} U$, and for $U \in \mathcal{B}_n$ put $S_n(U) = U - \bigcup_{i<n} V_i$. The sets $S_n(U)$ refine $\mathcal{A}$ and cover X: for x, take the smallest N with x in some $U \in \mathcal{B}_N$; then $x \in S_N(U)$. Local finiteness: with $x \in U \in \mathcal{B}_N$, the open set $U \subset V_N$ meets no $S_n(V)$ with $n > N$; for each $n \le N$ choose a neighbourhood W_n meeting finitely many elements of $\mathcal{B}_n$, hence finitely many $S_n(V)$. Then $U \cap W_1 \cap \cdots \cap W_N$ meets finitely many sets $S_n(V)$.

(2)⇒(3): let $\mathcal{B}$ be the open sets U with $\overline{U}$ contained in some element of $\mathcal{A}$; it covers X by regularity. Take a locally finite refinement $\mathcal{C}$ covering X; $\{\overline{C}\}$ is locally finite (Lemma 6.1), closed, covers X and refines $\mathcal{A}$.

(3)⇒(4): let $\mathcal{B}$ be a locally finite refinement of $\mathcal{A}$ covering X (by (3)). Let $\mathcal{C}$ be the open sets meeting only finitely many elements of $\mathcal{B}$; it covers X. By (3) take a closed locally finite refinement

$\mathcal{D}$ of $\mathcal{C}$ covering X; each $D \in \mathcal{D}$ meets only finitely many elements of $\mathcal{B}$. For $B \in \mathcal{B}$ let $C(B) = X - \bigcup\{D \in \mathcal{D} : D \cap B = \emptyset\}$, open by Lemma 6.1(c) and containing B, and choose $F(B) \in \mathcal{A}$ with $B \subset F(B)$. The collection $\mathcal{E} = \{C(B) \cap F(B) : B \in \mathcal{B}\}$ is an open refinement of $\mathcal{A}$ covering X. It is locally finite: a point has a neighbourhood W meeting only $D_1, \dots, D_k \in \mathcal{D}$, so $W \subset D_1 \cup \cdots \cup D_k$. If D meets $C(B)$, then D meets B (otherwise $D \subset X - C(B)$); hence each D_i meets only finitely many $C(B)$, and so does W. ■

Theorem 6.9 (41.4)

Every metrizable space is paracompact.

Proof. It is regular, and condition (1) of Lemma 6.8 holds by Lemma 6.2. ■

Theorem 6.10 (41.5)

Every regular Lindelöf space is paracompact.

Proof. A countable subcovering is a countable union of one-element (hence locally finite) collections, so condition (1) of Lemma 6.8 holds. ■

So $\mathbb{R}_\ell$ is paracompact, while $\mathbb{R}_\ell^2$ is not; and $\mathbb{R}^\omega$ in the product topology is paracompact (metrizable), whereas $\mathbb{R}^J$ for uncountable J is not (it is not normal).

Lemma 6.11 (41.6)

Let X be paracompact Hausdorff and $\{U_\alpha\}_{\alpha \in J}$ an indexed open covering. There is a locally finite indexed open covering $\{V_\alpha\}_{\alpha \in J}$ with $\overline{V_\alpha} \subset U_\alpha$ for each α (a **precise refinement**).

Proof. Let $\mathcal{A}$ be the open sets A with $\overline{A}$ in some U_α; it covers X by regularity (Theorem 6.6). Take a locally finite open refinement $\mathcal{B}$ covering X, and for each B choose $f(B) \in J$ with $\overline{B} \subset U_{f(B)}$. Let $V_\alpha = \bigcup\{B : f(B) = \alpha\}$. The V_α are open and cover X; a neighbourhood meeting finitely many B meets finitely many V_α; and $\overline{V_\alpha} = \bigcup_{f(B)=\alpha} \overline{B} \subset U_\alpha$ by Lemma 6.1(c). ■

Theorem 6.12 (41.7)

If X is paracompact Hausdorff and $\{U_\alpha\}_{\alpha \in J}$ is an indexed open covering, there is a partition of unity dominated by $\{U_\alpha\}$: continuous $\phi_\alpha : X \to [0,1]$ with $\operatorname{supp} \phi_\alpha \subset U_\alpha$, $\{\operatorname{supp} \phi_\alpha\}$ locally finite, and $\sum_\alpha \phi_\alpha = 1$.

Proof. By Lemma 6.11 twice, take locally finite open coverings $\{V_\alpha\}$, $\{W_\alpha\}$ with $\overline{W_\alpha} \subset V_\alpha$, $\overline{V_\alpha} \subset U_\alpha$. Urysohn (with X normal) gives ψ_α equal to 1 on $\overline{W_\alpha}$ and 0 off V_α, so $\operatorname{supp} \psi_\alpha \subset \overline{V_\alpha} \subset U_\alpha$, and these supports are locally finite (Lemma 6.1(b)). $\Psi = \sum \psi_\alpha$ is near each point a finite sum of continuous functions, hence continuous, and positive since the W_α cover. Put $\phi_\alpha = \psi_\alpha / \Psi$. ■

Theorem 6.13 (41.8)

Let X be paracompact Hausdorff, $\mathcal{C}$ a locally finite collection of subsets, and $\epsilon_C > 0$ for each $C \in \mathcal{C}$. There is a continuous $f : X \to \mathbb{R}$ with $f > 0$ everywhere and $f(x) \le \epsilon_C$ for $x \in C$.

Proof. Cover X by open sets each meeting finitely many elements of $\mathcal{C}$, and take a dominated partition of unity $\{\phi_U\}$ indexed by this covering. Let δ_U be the minimum of ϵ_C over the finitely many C meeting U ($\delta_U = 1$ if there are none), and $f = \sum_U \delta_U \phi_U$, continuous and positive. If $x \in C$ and $\phi_U(x) > 0$, then $x \in U$, so U meets C and $\delta_U \le \epsilon_C$; hence $f(x) \le \epsilon_C \sum \phi_U(x) = \epsilon_C$. ■

6.4 The Smirnov metrization theorem

Motivation. For paracompact spaces the metrization question reduces to a local one: a paracompact Hausdorff space is metrizable exactly when it is locally metrizable. This is how one shows that manifolds are metrizable.

X is **locally metrizable** if each point has a neighbourhood that is metrizable in the subspace topology.

Theorem 6.14 (42.1, Smirnov metrization theorem)

X is metrizable iff it is a paracompact Hausdorff space that is locally metrizable.

Proof. ($\Rightarrow$) Theorem 6.9.

($\Leftarrow$) X is regular (Theorem 6.6); we find a countably locally finite basis and apply Theorem 6.5. Cover X by open metrizable sets and take a locally finite open refinement $\mathcal{C}$ covering X; each $C \in \mathcal{C}$ has a metric d_C, and the balls $B_C(x,\epsilon) = \{y \in C : d_C(x,y) < \epsilon\}$ are open in X. For each m, let $\mathcal{A}_m$ be the covering by all balls $B_C(x, 1/m)$ ($C \in \mathcal{C}$, $x \in C$), and $\mathcal{D}_m$ a locally finite open refinement covering X. Then $\mathcal{D} = \bigcup \mathcal{D}_m$ is countably locally finite. Basis: let $x \in U$ open. Only finitely many $C_1, \dots, C_k \in \mathcal{C}$ contain x; choose ϵ_i with $B_{C_i}(x,\epsilon_i) \subset U \cap C_i$ and m with $2/m < \min \epsilon_i$. Choose $D \in \mathcal{D}_m$ with $x \in D$; then $D \subset B_C(y, 1/m)$ for some C and y, and $x \in C$, so $C = C_i$ for some i. By the triangle inequality $D \subset B_{C_i}(x, 2/m) \subset U$. ∎

Chapter 7. Complete Metric Spaces and Function Spaces

7.1 Complete metric spaces

Motivation. Completeness, every Cauchy sequence converges, is the property that makes limits exist without being given in advance. It is not a topological property (it depends on the metric), but function spaces with the uniform metric are complete, and every metric space can be completed.

A sequence (x_n) in (X,d) is **Cauchy** if for every $\epsilon>0$ there is N with $d(x_n,x_m)<\epsilon$ for $n,m\geq N$; (X,d) is **complete** if every Cauchy sequence converges. A closed subset of a complete space is complete, and a complete subspace of a metric space is closed. (X,d) is complete iff it is complete in the standard bounded metric $\bar{d}=\min(d,1)$, since the two have the same Cauchy sequences.

Lemma 7.1 (43.1)

A metric space is complete if every Cauchy sequence has a convergent subsequence.

Proof. Let (x_n) be Cauchy with $x_{n_i}\to x$. Given ϵ, choose N with $d(x_n,x_m)<\epsilon/2$ for $n,m\geq N$, and i with $n_i\geq N$ and $d(x_{n_i},x)<\epsilon/2$. Then $d(x_n,x)<\epsilon$ for $n\geq N$. ■

Theorem 7.2 (43.2)

$\mathbb{R}^k$ is complete in the euclidean metric d and in the square metric ρ.

Proof. Since $\rho\leq d\leq\sqrt{k}\rho$, the two metrics have the same Cauchy sequences; use ρ. A Cauchy sequence is bounded, so it lies in a cube $[-M,M]^k$, which is compact (Theorem 3.26), hence sequentially compact (Theorem 3.33). Apply Lemma 7.1. ■

Lemma 7.3 (43.3)

In $X=\prod X_\alpha$ (product topology), $x_n\to x$ iff $\pi_\alpha(x_n)\to\pi_\alpha(x)$ for each α.

Proof. ($\Rightarrow$) Each π_α is continuous.

($\Leftarrow$) A basis element $\prod U_\alpha$ containing x restricts only $\alpha_1,\ldots,\alpha_k$; choose N_i with $\pi_{\alpha_i}(x_n)\in U_{\alpha_i}$ for $n\geq N_i$, and let $N=\max N_i$. ■

Theorem 7.4 (43.4)

There is a metric on $\mathbb{R}^\omega$ (product topology) in which it is complete.

Proof. Take $D(x,y)=\sup_i \bar{d}(x_i,y_i)/i$, which induces the product topology (Theorem 2.36). If (x^n) is D-Cauchy, then $\bar{d}(x_i^n,x_i^m)\leq iD(x^n,x^m)$, so each coordinate sequence is Cauchy in $\mathbb{R}$ and converges. By Lemma 7.3, (x^n) converges in $\mathbb{R}^\omega$. ■

For a metric space (Y,d) and a set J, the **uniform metric** on Y^J is $\bar{\rho}(x,y)=\sup_\alpha \bar{d}(x_\alpha,y_\alpha)$.

Theorem 7.5 (43.5)

If Y is complete in d, then Y^J is complete in the uniform metric $\bar{\rho}$.

Proof. Y is complete in $\bar{d}$. If (f_n) is $\bar{\rho}$-Cauchy, then $\bar{d}(f_n(\alpha), f_m(\alpha)) \le \bar{\rho}(f_n, f_m)$, so each $f_n(\alpha)$ converges, say to $f(\alpha)$. Given ϵ, choose N with $\bar{\rho}(f_n, f_m) < \epsilon/2$ for $n, m \ge N$. For fixed $n \ge N$ and α, letting $m \to \infty$ gives $\bar{d}(f_n(\alpha), f(\alpha)) \le \epsilon/2$; hence $\bar{\rho}(f_n, f) \le \epsilon/2 < \epsilon$. ■

Let $\mathcal{C}(X, Y)$ be the continuous and $\mathcal{B}(X, Y)$ the bounded functions $X \to Y$. On $\mathcal{B}(X, Y)$ the **sup metric** is $\rho(f, g) = \sup_x d(f(x), g(x))$; there $\bar{\rho} = \min(\rho, 1)$.

Theorem 7.6 (43.6)

Let X be a space and (Y, d) a metric space. $\mathcal{C}(X, Y)$ and $\mathcal{B}(X, Y)$ are closed in Y^X under the uniform metric. Hence if Y is complete, they are complete in $\bar{\rho}$, and $\mathcal{B}(X, Y)$ is complete in the sup metric ρ.

Proof. If $f_n \to f$ in $\bar{\rho}$ with f_n continuous, then $f_n \to f$ uniformly (for $\epsilon < 1$, $\bar{\rho} < \epsilon$ means $d < \epsilon$ everywhere), so f is continuous (Theorem 2.42). If the f_n are bounded, choose N with $\bar{\rho}(f_N, f) < \frac{1}{2}$; then $d(f_N(x), f(x)) < \frac{1}{2}$ for all x, so $\operatorname{diam} f(X) \le \operatorname{diam} f_N(X) + 1$. Completeness follows from Theorem 7.5, and for ρ because ρ and $\bar{\rho} = \min(\rho, 1)$ have the same Cauchy and convergent sequences. ■

Theorem 7.7 (43.7)

Every metric space (X, d) has an isometric imbedding into a complete metric space.

Proof. $\mathcal{B}(X, \mathbb{R})$ is complete in the sup metric (Theorem 7.6). Fix x_0 and for $a \in X$ define $\phi_a(x) = d(x, a) - d(x, x_0)$; it is bounded by $d(a, x_0)$. Now $|\phi_a(x) - \phi_b(x)| = |d(x, a) - d(x, b)| \le d(a, b)$, with equality at $x = a$; so $\rho(\phi_a, \phi_b) = d(a, b)$ and $a \mapsto \phi_a$ is an isometric imbedding. ■

The closure of the image is a **completion** of X: a complete metric space containing an isometric copy of X as a dense subset. It is unique up to an isometry fixing X. Completeness is not a topological property: $\mathbb{R}$ is complete but the homeomorphic $(-1, 1)$ is not.

7.2 A space-filling curve

Motivation. Completeness of the space of continuous functions can be used to construct surprising objects. Here a uniform limit of polygonal paths gives a continuous surjection of $[0, 1]$ onto the square, showing that dimension is not preserved by continuous maps.

Theorem 7.8 (44.1)

Let $I = [0, 1]$. There is a continuous map $f : I \to I^2$ whose image is all of I^2.

Proof.

Step 1. call a path in a square $[a, a + s] \times [b, b + s]$ *triangular* if it runs linearly from the lower left corner to the centre of the top edge and then to the lower right corner. Given a triangular path in a square, divide the square into four subsquares of side $s/2$ and replace the path by the concatenation of four triangular paths, one in each subsquare, traversed in the order lower left, upper left, upper right, lower right (with orientations chosen so that the path still starts and ends at the original endpoints). Applied to a path made of many triangular paths, the operation is performed on each one.

Step 2. let f_0 be the triangular path in I^2 and f_n the result of applying the operation n times. Then I is divided into 4^n intervals of length 4^{-n}, and on each such interval f_n is a triangular path

in a subsquare of side 2^{-n}; the 4^n subsquares are exactly the squares of the grid of side 2^{-n}. On the same interval f_{n+1} stays in the same subsquare. Hence $d(f_n(t), f_{n+1}(t)) \le \sqrt{2}\,2^{-n}$ for all t, and (f_n) is Cauchy in the sup metric on $\mathcal{C}(I, I^2)$, which is complete (Theorem 7.6, I^2 being closed in the complete $\mathbb{R}^2$). Let f be the limit; it is continuous, and $d(f(t), f_n(t)) \le \sum_{k\ge n} \sqrt{2}\,2^{-k} = 2\sqrt{2}\,2^{-n}$.

Step 3. given $x \in I^2$, x lies in some grid square of side 2^{-n}, which f_n visits, so there is t_n with $d(f_n(t_n), x) \le \sqrt{2}\,2^{-n}$. Then $d(f(t_n), x) \le 3\sqrt{2}\,2^{-n} \to 0$, so $x \in \overline{f(I)} = f(I)$, the image being compact and hence closed. ■

7.3 Compactness in metric spaces

Motivation. In metric spaces compactness splits into two simpler properties: completeness and total boundedness. This characterization is what we will use to find compact subsets of function spaces.

(X, d) is **totally bounded** if for each $\epsilon > 0$ it is covered by finitely many ϵ-balls.

Theorem 7.9 (45.1)

A metric space is compact iff it is complete and totally bounded.

Proof. ($\Rightarrow$) A Cauchy sequence has a convergent subsequence (Theorem 3.33), hence converges (Lemma 7.1); and finitely many of the ϵ-balls cover.

($\Leftarrow$) We show sequential compactness. Let (x_n) be given. Cover X by finitely many balls of radius 1; one, B_1, contains x_n for n in an infinite set J_1. Given an infinite J_{k-1}, cover X by finitely many balls of radius $1/k$; one, B_k, contains x_n for infinitely many $n \in J_{k-1}$; call them J_k. Choose $n_1 < n_2 < \cdots$ with $n_k \in J_k$. For $i, j \ge k$, $x_{n_i}, x_{n_j} \in B_k$, so $d(x_{n_i}, x_{n_j}) < 2/k$. The subsequence is Cauchy and converges. ■

A family $\mathcal{F} \subset \mathcal{C}(X, Y)$ is **equicontinuous at** x_0 if for each $\epsilon > 0$ there is a neighbourhood U of x_0 with $d(f(x), f(x_0)) < \epsilon$ for all $x \in U$ and all $f \in \mathcal{F}$; it is **equicontinuous** if this holds at every point.

Lemma 7.10 (45.2)

Let X be a space and (Y, d) a metric space. If $\mathcal{F} \subset \mathcal{C}(X, Y)$ is totally bounded under the uniform metric, then $\mathcal{F}$ is equicontinuous under d.

Proof. Let $0 < \epsilon < 1$ and $x_0 \in X$. Cover $\mathcal{F}$ by $\bar\rho$-balls $B(f_1, \epsilon/3), \ldots, B(f_n, \epsilon/3)$, and choose a neighbourhood U of x_0 with $d(f_i(x), f_i(x_0)) < \epsilon/3$ for all i and $x \in U$. For $f \in \mathcal{F}$ pick i with $\bar\rho(f, f_i) < \epsilon/3$; since $\epsilon/3 < 1$, $d(f(x), f_i(x)) < \epsilon/3$ for all x. For $x \in U$ the triangle inequality gives $d(f(x), f(x_0)) < \epsilon$. ■

Lemma 7.11 (45.3)

Let X and Y be compact, (Y, d) metric. If $\mathcal{F} \subset \mathcal{C}(X, Y)$ is equicontinuous under d, then $\mathcal{F}$ is totally bounded under the sup metric and the uniform metric.

Proof. Since $\bar\rho \le \rho$, it suffices to treat ρ. Let $\epsilon > 0$. For each $a \in X$ choose a neighbourhood U_a with $d(f(x), f(a)) < \epsilon/3$ for $x \in U_a$, $f \in \mathcal{F}$; finitely many $U_{a_1}, \ldots, U_{a_k}$ cover X. Cover Y by finitely many open $V_1, \ldots, V_m$ of diameter less than $\epsilon/3$. For each function $\alpha : \{1, \ldots, k\} \to \{1, \ldots, m\}$ for which some $f \in \mathcal{F}$ has $f(a_i) \in V_{\alpha(i)}$ for all i, choose one such f_α. Given $f \in \mathcal{F}$, take α with

$f(a_i) \in V_{\alpha(i)}$ for all i. For $x \in U_{a_i}$,

$$d(f(x), f_\alpha(x)) \le d(f(x), f(a_i)) + d(f(a_i), f_\alpha(a_i)) + d(f_\alpha(a_i), f_\alpha(x)) < \epsilon.$$

The continuous function $d(f(x), f_\alpha(x))$ attains its maximum on the compact X, so $\rho(f, f_\alpha) < \epsilon$. The finitely many balls $B_\rho(f_\alpha, \epsilon)$ cover $\mathcal{F}$. ■

$\mathcal{F} \subset \mathcal{C}(X, \mathbb{R}^n)$ is **pointwise bounded** if $\mathcal{F}_a = \{f(a) : f \in \mathcal{F}\}$ is bounded for each a.

Theorem 7.12 (45.4, Ascoli's theorem, classical version)

Let X be compact and give $\mathcal{C}(X, \mathbb{R}^n)$ the uniform topology (d the square or the euclidean metric). A subspace $\mathcal{F}$ has compact closure iff $\mathcal{F}$ is equicontinuous and pointwise bounded under d.

Proof. Let $\mathcal{G} = \overline{\mathcal{F}}$; on $\mathcal{C}(X, \mathbb{R}^n)$ the sup metric ρ induces the uniform topology.

($\Rightarrow$) $\mathcal{G}$ is compact, hence totally bounded under ρ, and so is $\mathcal{F}$; by Lemma 7.10, $\mathcal{F}$ is equicontinuous. Evaluation $f \mapsto f(a)$ is continuous, so $\mathcal{G}_a$ is compact, hence bounded, and $\mathcal{F}_a \subset \mathcal{G}_a$.

($\Leftarrow$)

Step 1. $\mathcal{G}$ is equicontinuous and pointwise bounded. Given a and ϵ, choose U with $d(f(x), f(a)) < \epsilon/3$ for $x \in U$, $f \in \mathcal{F}$; for $g \in \mathcal{G}$ choose $f \in \mathcal{F}$ with $\rho(f, g) < \epsilon/3$; then $d(g(x), g(a)) < \epsilon$ on U. Also each $g(a)$ is within 1 of $\mathcal{F}_a$, so $\mathcal{G}_a$ is bounded.

Step 2. the union of the images $g(X)$, $g \in \mathcal{G}$, lies in a compact $Y \subset \mathbb{R}^n$. Choose neighbourhoods U_a with $d(g(x), g(a)) < 1$ for $x \in U_a$, $g \in \mathcal{G}$, and cover X by $U_{a_1}, \dots, U_{a_k}$. Every $g(x)$ is within 1 of the bounded set $\mathcal{G}_{a_1} \cup \cdots \cup \mathcal{G}_{a_k}$, so all images lie in a closed bounded cube Y, compact by Theorem 3.26.

Step 3. $\mathcal{G} \subset \mathcal{C}(X, Y)$ is closed in the complete space $\mathcal{C}(X, \mathbb{R}^n)$ (Theorems 7.2, 7.6), hence complete; it is totally bounded by Lemma 7.11; hence compact by Theorem 7.9. ■

Corollary 7.13 (45.5)

Let X be compact. A subspace $\mathcal{F}$ of $\mathcal{C}(X, \mathbb{R}^n)$ (uniform topology) is compact iff it is closed, bounded under the sup metric ρ, and equicontinuous under d.

Proof. A compact $\mathcal{F}$ is closed and bounded in the metric space, and equicontinuous by Theorem 7.12. Conversely, if $\mathcal{F}$ is closed it equals its closure; ρ-boundedness implies pointwise boundedness; apply Theorem 7.12. ■

Example 7.1

Let $f_n : [0, 1] \to \mathbb{R}$ be differentiable with $|f_n(0)| \le 1$ and $|f_n'| \le 1$. By the mean value theorem $|f_n(x) - f_n(y)| \le |x - y|$ (equicontinuity) and $|f_n| \le 2$ (pointwise boundedness), so some subsequence converges uniformly. The family $\{x^n\}$ on $[0, 1]$ is bounded but not equicontinuous at 1, and indeed has no uniformly convergent subsequence.

7.4 Pointwise and compact convergence

Motivation. Different modes of convergence of functions correspond to different topologies on function spaces. Uniform convergence is often too strong and pointwise too weak; convergence on compact sets is the one that preserves continuity and fits analysis.

For a set X and space Y, the **topology of pointwise convergence** on Y^X has subbasis $S(x,U) = \{f : f(x) \in U\}$ ($x \in X$, U open in Y); it is the product topology.

Theorem 7.14 (46.1)

$f_n \to f$ in the topology of pointwise convergence iff $f_n(x) \to f(x)$ for each $x \in X$.

Proof. This is Lemma 7.3 for the product topology. ■

Let X be a space and (Y,d) a metric space. For $f \in Y^X$, compact $C \subset X$ and $\epsilon > 0$ let $B_C(f,\epsilon) = \{g : \sup_{x\in C} d(f(x),g(x)) < \epsilon\}$. These sets form a basis: if $g \in B_C(f,\epsilon)$ and $\delta = \epsilon - \sup_C d(f,g)$, then $B_C(g,\delta) \subset B_C(f,\epsilon)$, and if g lies in $B_{C_1}(f_1,\epsilon_1) \cap B_{C_2}(f_2,\epsilon_2)$ then, with the corresponding δ_1, δ_2, $B_{C_1\cup C_2}(g, \min \delta_i)$ lies in both. The resulting topology is the **topology of compact convergence**.

Theorem 7.15 (46.2)

$f_n \to f$ in the topology of compact convergence iff for each compact $C \subset X$, $f_n|C \to f|C$ uniformly.

Proof. Every basis element containing f contains some $B_C(f,\epsilon)$, so convergence means: for every C and ϵ, eventually $\sup_C d(f_n,f) < \epsilon$. That is uniform convergence on each C. ■

X is **compactly generated** if $A \subset X$ is open whenever $A \cap C$ is open in C for every compact C (equivalently for closed sets).

Lemma 7.16 (46.3)

If X is locally compact, or first countable, then X is compactly generated.

Proof. Suppose $A \cap C$ is open in C for every compact C, and $x \in A$. If X is locally compact, take a compact C containing a neighbourhood U of x; then $A \cap U$ is open in U, hence in X, and $x \in A \cap U \subset A$. If X is first countable and A is not open, some $x \in A$ has no neighbourhood inside A; with a nested countable basis $B_1 \supset B_2 \supset \cdots$ at x, pick $x_n \in B_n - A$. Then $x_n \to x$, and $C = \{x\} \cup \{x_n\}$ is compact. The set $A \cap C$ is open in C and contains x, so it contains x_n for large n, a contradiction. ■

Lemma 7.17 (46.4)

If X is compactly generated, then $f : X \to Y$ is continuous whenever $f|C$ is continuous for every compact $C \subset X$.

Proof. For open $V \subset Y$, $f^{-1}(V) \cap C = (f|C)^{-1}(V)$ is open in C for every C, so $f^{-1}(V)$ is open. ■

Theorem 7.18 (46.5)

Let X be compactly generated and (Y,d) metric. Then $\mathcal{C}(X,Y)$ is closed in Y^X in the topology of compact convergence.

Proof. Let f be in the closure of $\mathcal{C}(X,Y)$ and C compact. For each n choose $f_n \in B_C(f,1/n) \cap \mathcal{C}(X,Y)$. Then $f_n|C \to f|C$ uniformly, so $f|C$ is continuous (Theorem 2.42). By Lemma 7.17, f is continuous. ■

Corollary 7.19 (46.6)

If X is compactly generated, (Y,d) metric, and continuous f_n converge to f in the topology of compact convergence, then f is continuous.

Proof. The limit of a sequence in a set lies in its closure; apply Theorem 7.18. ■

Theorem 7.20 (46.7)

For Y^X with (Y,d) metric: (uniform) $\supset$ (compact convergence) $\supset$ (pointwise convergence). If X is compact the first two coincide; if X is discrete the last two coincide.

Proof. If $g \in B_C(f,\epsilon)$ and $\delta = \epsilon - \sup_C d(f,g)$, the uniform ball $B_{\bar\rho}(g, \min(\delta,1))$ lies in $B_C(f,\epsilon)$, because $\bar\rho(g,h) < \min(\delta,1)$ forces $d(g(x),h(x)) < \delta$ for all x. If $g \in S(x,U)$, choose ϵ with $B(g(x),\epsilon) \subset U$; then $B_{\{x\}}(g,\epsilon) \subset S(x,U)$. If X is compact, for $\epsilon \le 1$ the uniform ball $B_{\bar\rho}(f,\epsilon)$ is exactly $B_X(f,\epsilon)$. If X is discrete, compact sets are finite and $B_C(f,\epsilon) = \bigcap_{x\in C} S(x, B(f(x),\epsilon))$. ■

For spaces X, Y, the **compact-open topology** on $\mathcal{C}(X,Y)$ has subbasis $S(C,U) = \{f : f(C) \subset U\}$ with C compact and U open.

Theorem 7.21 (46.8)

If (Y,d) is metric, the compact-open topology on $\mathcal{C}(X,Y)$ equals the topology of compact convergence.

Proof. Let $f \in S(C,U)$. The continuous function $y \mapsto d(y, Y-U)$ is positive on the compact $f(C)$ (if $U \neq Y$), so has a positive minimum ϵ; then $B_C(f,\epsilon) \subset S(C,U)$. Conversely consider a basic set $B_C(f,\epsilon)$ with f continuous (it suffices to treat those centred at f). For each $x \in C$ choose a neighbourhood V_x with $f(V_x) \subset B(f(x),\epsilon/4)$, so $f(\overline{V_x}) \subset B(f(x),\epsilon/3)$. Cover C by $V_{x_1},\dots,V_{x_n}$ and put $C_i = \overline{V_{x_i}} \cap C$ (compact), $U_i = B(f(x_i),\epsilon/3)$. Then $f \in W = \bigcap S(C_i,U_i)$. If $g \in W$ and $x \in C_i$, both $g(x)$ and $f(x)$ lie in U_i, so $d(f(x),g(x)) < 2\epsilon/3$. Hence $\sup_C d(f,g) \le 2\epsilon/3 < \epsilon$ and $W \subset B_C(f,\epsilon)$. ■

Corollary 7.22 (46.9)

For metric Y, the topology of compact convergence on $\mathcal{C}(X,Y)$ does not depend on the metric of Y; hence if X is compact, neither does the uniform topology on $\mathcal{C}(X,Y)$.

Proof. The compact-open topology is defined without a metric; use Theorems 7.21 and 7.20. ■

Theorem 7.23 (46.10)

If X is locally compact Hausdorff and $\mathcal{C}(X,Y)$ has the compact-open topology, the **evaluation map** $e : X \times \mathcal{C}(X,Y) \to Y$, $e(x,f) = f(x)$, is continuous.

Proof. Let $f(x) \in V$, V open. By Theorem 3.35 there is a neighbourhood U of x with $\overline{U}$ compact and $\overline{U} \subset f^{-1}(V)$. Then $U \times S(\overline{U},V)$ is a neighbourhood of (x,f) that e maps into V. ■

Theorem 7.24 (46.11)

Give $\mathcal{C}(X,Y)$ the compact-open topology. If $f : X \times Z \to Y$ is continuous, so is the induced map $F : Z \to \mathcal{C}(X,Y)$, $F(z)(x) = f(x,z)$. The converse holds if X is locally compact Hausdorff.

Proof. Each $F(z)$ is continuous, as a composite $x \mapsto (x,z) \mapsto f(x,z)$. Let $F(z_0) \in S(C,U)$; then $C \times \{z_0\} \subset f^{-1}(U)$, and since C is compact the tube lemma (Lemma 3.21) gives a neighbourhood W of z_0 with $C \times W \subset f^{-1}(U)$; thus $F(W) \subset S(C,U)$. Conversely, $f = e \circ (\mathrm{id}_X \times F)$ is continuous by Theorem 7.23. ∎

Thus, for locally compact Hausdorff X, a homotopy $X \times I \to Y$ is the same as a path in $\mathcal{C}(X,Y)$.

7.5 Ascoli's theorem

Motivation. Which families of continuous functions have uniformly convergent subsequences? Ascoli's theorem answers: those that are equicontinuous and pointwise bounded. It is the function-space version of Bolzano–Weierstrass and is used to prove existence theorems for differential equations.

Theorem 7.25 (47.1, Ascoli's theorem)

Let X be a space, (Y,d) a metric space, and $\mathcal{C}(X,Y)$ with the topology of compact convergence; let $\mathcal{F} \subset \mathcal{C}(X,Y)$.
(a) If $\mathcal{F}$ is equicontinuous under d and each $\mathcal{F}_a = \{f(a) : f \in \mathcal{F}\}$ has compact closure, then $\mathcal{F}$ lies in a compact subspace of $\mathcal{C}(X,Y)$.
(b) The converse holds if X is locally compact Hausdorff.

Proof. (a) Let $\mathcal{G}$ be the closure of $\mathcal{F}$ in Y^X with the product topology.

Step 1. $\mathcal{G}$ is equicontinuous and $\mathcal{G}_a \subset \overline{\mathcal{F}_a}$. Given a and ϵ, take U with $d(f(x),f(a)) < \epsilon/3$ for $x \in U$, $f \in \mathcal{F}$. For $g \in \mathcal{G}$ and $x \in U$, the product neighbourhood $\{h : h(x) \in B(g(x),\epsilon/3),\ h(a) \in B(g(a),\epsilon/3)\}$ of g contains some $f \in \mathcal{F}$, whence $d(g(x),g(a)) < \epsilon$. In particular $\mathcal{G} \subset \mathcal{C}(X,Y)$. Since π_a is continuous, $\pi_a(\mathcal{G}) \subset \overline{\pi_a(\mathcal{F})}$.

Step 2. $\mathcal{G}$ is compact in the product topology, being closed in $\prod_a \overline{\mathcal{F}_a}$, which is compact by Tychonoff (Theorem 5.3).

Step 3. on $\mathcal{G}$ the product topology and the topology of compact convergence agree. One is coarser (Theorem 7.20); for the other, let $g \in \mathcal{G}$, C compact, $\epsilon > 0$. For $x \in C$ choose U_x with $d(h(y),h(x)) < \epsilon/3$ for $y \in U_x$, $h \in \mathcal{G}$; cover C by $U_{x_1},\dots,U_{x_n}$, and let $W = \{h \in \mathcal{G} : d(h(x_i),g(x_i)) < \epsilon/3\ \forall i\}$, open in the product topology. For $h \in W$ and $y \in C \cap U_{x_i}$, $d(h(y),g(y)) < \epsilon$ by three applications of the triangle inequality; the maximum over compact C is $< \epsilon$, so $W \subset B_C(g,\epsilon)$. Therefore $\mathcal{G}$ is a compact subspace of $\mathcal{C}(X,Y)$ containing $\mathcal{F}$.

(b) Let $\mathcal{H} \supset \mathcal{F}$ be compact in $\mathcal{C}(X,Y)$. Evaluation at a is continuous (the pointwise topology is coarser), so $\mathcal{H}_a$ is compact and $\overline{\mathcal{F}_a} \subset \mathcal{H}_a$ is compact. For equicontinuity at a, choose a compact C containing a neighbourhood of a. Restriction $r : \mathcal{C}(X,Y) \to \mathcal{C}(C,Y)$ is continuous when $\mathcal{C}(C,Y)$ has the uniform topology, since $r^{-1}(B_C(r(f),\epsilon)) \supset B_C(f,\epsilon)$ and for compact C these balls give the uniform topology. So $r(\mathcal{H})$ is compact, hence totally bounded under $\bar\rho$, hence equicontinuous on C (Lemma 7.10). As C contains a neighbourhood of a, $\mathcal{H}$, and therefore $\mathcal{F}$, is equicontinuous at a. ∎

For $Y = \mathbb{R}^n$, the condition on $\mathcal{F}_a$ is pointwise boundedness, and for compact X the theorem reduces to Theorem 7.12.

Chapter 8. Baire Spaces and Dimension Theory

8.1 Baire spaces

Motivation. A complete metric space cannot be a countable union of nowhere-dense sets. This simple statement, the Baire category theorem, is a powerful existence tool: to show that some object exists, show that the set of "bad" objects is small in the sense of category.

$A \subset X$ has **empty interior** if it contains no nonempty open set. X is a **Baire space** if every countable union of closed sets with empty interior has empty interior.

Example 8.1

$\mathbb{Q}$ is not a Baire space: it is the countable union of its one-point sets, which are closed with empty interior. $\mathbb{Z}_+$ is a Baire space: every nonempty subset is open, so the only set with empty interior is $\emptyset$. The irrationals form a Baire space, since they admit a complete metric (they are homeomorphic to $\mathbb{Z}_+^\omega$); we do not prove this here.

Lemma 8.1 (48.1)

X is a Baire space iff for every countable collection $\{U_n\}$ of dense open sets, $\bigcap U_n$ is dense.

Proof. U is open and dense iff $X - U$ is closed with empty interior, and $\bigcap U_n$ is dense iff $X - \bigcap U_n = \bigcup (X - U_n)$ has empty interior. ∎

Theorem 8.2 (48.2, Baire category theorem)

If X is a compact Hausdorff space or a complete metric space, then X is a Baire space.

Proof. Let A_n be closed with empty interior and U_0 nonempty open; we find a point of U_0 outside every A_n. Since $U_0 \not\subset A_1$, pick $y \in U_0 - A_1$. X is regular (Theorem 4.8, or metric), so there is a neighbourhood U_1 of y with $\overline{U_1} \cap A_1 = \emptyset$ and $\overline{U_1} \subset U_0$; in the metric case choose also $\operatorname{diam} U_1 < 1$. Inductively, given nonempty open U_{n-1}, choose nonempty open U_n with $\overline{U_n} \subset U_{n-1}$, $\overline{U_n} \cap A_n = \emptyset$, and (metric case) $\operatorname{diam} U_n < 1/n$. The nested nonempty closed sets $\overline{U_n}$ have nonempty intersection: by the finite intersection property in the compact case, and by Lemma 8.3 in the complete case. A point of the intersection lies in U_0 and in no A_n. ∎

Lemma 8.3 (48.3)

If $C_1 \supset C_2 \supset \cdots$ are nonempty closed sets in a complete metric space with $\operatorname{diam} C_n \to 0$, then $\bigcap C_n \neq \emptyset$.

Proof. Pick $x_n \in C_n$. For $n, m \geq N$, both lie in C_N, so $d(x_n, x_m) \leq \operatorname{diam} C_N$; the sequence is Cauchy and converges to some x. For each k, $x_n \in C_k$ for $n \geq k$, so $x \in C_k$. ∎

Lemma 8.4 (48.4)

An open subspace Y of a Baire space X is a Baire space.

Proof. Let A_n be closed in Y with empty interior in Y, and $\overline{A_n}$ its closure in X, so $\overline{A_n} \cap Y = A_n$. If U is a nonempty open subset of X contained in $\overline{A_n}$, then U meets $A_n \subset Y$, and $U \cap Y$ is a

nonempty open subset of Y contained in A_n, impossible. So $\overline{A_n}$ has empty interior in X. If U is open in Y (hence in X) and $U \subset \bigcup A_n \subset \bigcup \overline{A_n}$, then $U = \emptyset$ because X is Baire. ∎

Theorem 8.5 (48.5)

Let X be a Baire space, (Y, d) metric, and $f_n : X \to Y$ continuous with $f_n(x) \to f(x)$ for every x. Then the set of points where f is continuous is dense in X.

Proof. For $N \in \mathbb{Z}_+$ and $\epsilon > 0$ let $A_N(\epsilon) = \{x : d(f_n(x), f_m(x)) \le \epsilon \ \forall n, m \ge N\}$, closed as an intersection of closed sets.

Step 1. every nonempty open U contains a nonempty open V such that, for some N, $d(f_n(x), f(x)) \le \epsilon$ for all $x \in V$, $n \ge N$. Indeed, since $(f_n(x))$ is Cauchy for each x, $X = \bigcup_N A_N(\epsilon)$; U is Baire (Lemma 8.4), so some $A_N(\epsilon) \cap U$ has nonempty interior V in U, and V is open in X. Letting $m \to \infty$ in the defining inequality gives the claim.

Step 2. let G_M be the union of all open V as in Step 1 for $\epsilon = 1/M$. It is open and, by Step 1, dense. So $C = \bigcap_M G_M$ is dense (Lemma 8.1). Let $x_0 \in C$ and $\epsilon > 0$; choose M with $1/M < \epsilon/3$, an open $V \ni x_0$ and N with $d(f_n(x), f(x)) \le 1/M$ on V for $n \ge N$, and a neighbourhood $W \subset V$ of x_0 with $d(f_N(x), f_N(x_0)) < \epsilon/3$ on W. Then for $x \in W$,

$$d(f(x), f(x_0)) \le d(f(x), f_N(x)) + d(f_N(x), f_N(x_0)) + d(f_N(x_0), f(x_0)) < \epsilon.$$

So f is continuous at each point of the dense set C. ∎

8.2 A nowhere-differentiable function

Motivation. As an application of Baire's theorem, most continuous functions (in the sense of category) are nowhere differentiable, even though no explicit example is needed in the proof.

Theorem 8.6 (49.1)

Let $h : [0,1] \to \mathbb{R}$ be continuous and $\epsilon > 0$. There is a continuous, nowhere differentiable $g : [0,1] \to \mathbb{R}$ with $|h(x) - g(x)| < \epsilon$ for all x.

Proof. Let $I = [0,1]$ and $\mathcal{C} = \mathcal{C}(I, \mathbb{R})$ with the sup metric ρ; it is complete (Theorem 7.6), hence Baire. For $f \in \mathcal{C}$, $x \in I$ and $0 < k \le \frac{1}{2}$ let

$$\Delta f(x,k) = \max\Big\{\Big|\frac{f(x+k) - f(x)}{k}\Big|, \Big|\frac{f(x-k) - f(x)}{k}\Big|\Big\},$$

the maximum taken over those of $x \pm k$ lying in I (at least one does), and $\|\Delta_k f\| = \inf_x \Delta f(x,k)$. Let U_n be the set of f for which $\|\Delta_k f\| > n$ for some $k \le 1/n$.

(a) $\bigcap U_n$ consists of nowhere differentiable functions. If $f \in \bigcap U_n$, there are $k_n \le 1/n$ with $\Delta f(x, k_n) > n$ for all x. If f were differentiable at x (one-sidedly at an endpoint), the difference quotients would tend to $f'(x)$, so $\Delta f(x, k_n) \to |f'(x)|$, contradicting $\Delta f(x, k_n) > n$.

(b) U_n is open. Let $f \in U_n$ with $M = \|\Delta_k f\| > n$, and $\delta = (M - n)k/4$. If $\rho(f, g) < \delta$, then $|g(x \pm k) - g(x)| \ge |f(x \pm k) - f(x)| - 2\delta$, so $\Delta g(x,k) \ge \Delta f(x,k) - (M-n)/2$ and $\|\Delta_k g\| \ge (M+n)/2 > n$.

(c) U_n is dense. Let $f \in \mathcal{C}$ and $\epsilon' > 0$. By uniform continuity choose a partition of I on whose subintervals f varies by less than $\epsilon'/2$; the piecewise linear p interpolating f at the partition points satisfies $\rho(f, p) < \epsilon'/2$ ($p(x)$ is a convex combination of two values each within $\epsilon'/2$ of $f(x)$). Let L

bound the slopes of p, so $|p(x)-p(y)| \le L|x-y|$. Choose m with $r = m\epsilon'/4 \ge L+n+1$, and let ϕ be the sawtooth that is linear on each $[j/m,(j+1)/m]$, alternately rising from 0 to $\epsilon'/4$ and falling back, so $0 \le \phi \le \epsilon'/4$ and ϕ has slope $\pm r$. Put $g = p+\phi$, so $\rho(f,g) < \epsilon'$, and $k = \min(1/n, 1/2m)$. Each x lies in some $[j/m,(j+1)/m]$ of length $1/m \ge 2k$, so $x+k$ or $x-k$ lies in the same subinterval, where $|\phi(x\pm k)-\phi(x)| = rk$. Then $|g(x\pm k)-g(x)| \ge rk - Lk$, so $\Delta g(x,k) \ge r - L \ge n+1$ for all x, and $g \in U_n$.

By Lemma 8.1, $\bigcap U_n$ is dense, so it contains some g with $\rho(h,g) < \epsilon$. ■

8.3 Introduction to dimension theory

Motivation. What does it mean for a space to have dimension m? The covering definition says: every open cover can be refined so that no point lies in more than $m+1$ sets. With it we show that every compact m-manifold embeds in $\mathbb{R}^{2m+1}$.

A collection $\mathcal{A}$ has **order** $m+1$ if some point lies in $m+1$ elements and no point lies in more than $m+1$. X is **finite dimensional** if for some m every open covering has an open refinement covering X of order at most $m+1$; the least such m is the **topological dimension** $\dim X$. It is a topological invariant. Every compact subspace of $\mathbb{R}$ has dimension at most 1; an arc has dimension 1.

Theorem 8.7 (50.1)

If X is finite dimensional and $Y \subset X$ is closed, then Y is finite dimensional and $\dim Y \le \dim X$.

Proof. Let $m = \dim X$ and $\mathcal{A}$ a covering of Y by sets open in Y; choose A' open in X with $A' \cap Y = A$. Cover X by the A' and $X - Y$, take an open refinement $\mathcal{B}$ of order at most $m+1$ covering X, and intersect its elements with Y: this refines $\mathcal{A}$, covers Y, and has order at most $m+1$. ■

Theorem 8.8 (50.2)

If $X = Y \cup Z$ with Y, Z closed and finite dimensional, then $\dim X = \max\{\dim Y, \dim Z\}$.

Proof. Let $m = \max\{\dim Y, \dim Z\}$; $\dim X \ge m$ by Theorem 8.7.

Step 1. every open covering $\mathcal{A}$ of X has an open refinement covering X in which each point of Y lies in at most $m+1$ elements. Take a refinement $\mathcal{C}$ of $\{A \cap Y\}$ by sets open in Y, covering Y, of order at most $m+1$. For $C \in \mathcal{C}$ choose U_C open in X with $U_C \cap Y = C$ and $A_C \in \mathcal{A}$ with $C \subset A_C$. The collection $\{U_C \cap A_C\} \cup \{A - Y : A \in \mathcal{A}\}$ works: it covers X, refines $\mathcal{A}$, is open since Y is closed, and a point $y \in Y$ in $U_C \cap A_C$ lies in C.

Step 2. given $\mathcal{A}$, apply Step 1 to get $\mathcal{B}$ (good at points of Y), then Step 1 for Z to $\mathcal{B}$ to get $\mathcal{D}$ (good at points of Z). Choose $f : \mathcal{D} \to \mathcal{B}$ with $D \subset f(D)$ and let $E(B) = \bigcup\{D : f(D) = B\}$. The sets $E(B)$ are open, cover X and refine $\mathcal{A}$. If x lies in distinct $E(B_1), \dots, E(B_k)$, then x lies in distinct $B_1, \dots, B_k$ and in distinct $D_1, \dots, D_k$ with $f(D_i) = B_i$; so $k \le m+1$ whether $x \in Y$ or $x \in Z$. ■

Corollary 8.9 (50.3)

If $X = Y_1 \cup \cdots \cup Y_k$ with each Y_i closed and finite dimensional, then $\dim X = \max\{\dim Y_1, \dots, \dim Y_k\}$.

Proof. Induction on k using Theorem 8.8. ■

Points $x_0, \dots, x_k$ of $\mathbb{R}^N$ are **geometrically independent** if $\sum a_i x_i = 0$ and $\sum a_i = 0$ imply all $a_i = 0$; equivalently, no x_j lies in the plane P spanned by the others, $P = \{\sum t_i x_i : \sum t_i = 1\}$. A set of points of $\mathbb{R}^N$ is in **general position** if each subset of at most $N+1$ of its points is geometrically independent.

Lemma 8.10 (50.4)

Given $x_1, \dots, x_n \in \mathbb{R}^N$ and $\delta > 0$, there are $y_1, \dots, y_n$ in general position with $|x_i - y_i| < \delta$.

Proof. Induction: let $y_1 = x_1$, and suppose $y_1, \dots, y_p$ are in general position. Each subset T of at most N of them spans a plane $P(T)$ of dimension at most $N-1$, a closed set with empty interior. A finite union of closed sets with empty interior has empty interior (if $U \subset A \cup B$ is open and nonempty, then $U - A$ is a nonempty open subset of B). So choose y_{p+1} within δ of x_{p+1} outside every $P(T)$. A subset S of at most $N+1$ of $y_1, \dots, y_{p+1}$ is independent: either $S \subset \{y_1, \dots, y_p\}$, or $S = T \cup \{y_{p+1}\}$; then if $\sum_T a_i y_i + a y_{p+1} = 0$ with $\sum a_i + a = 0$ and $a \neq 0$, $y_{p+1} = \sum(-a_i/a) y_i \in P(T)$, a contradiction; so $a = 0$ and all $a_i = 0$. ∎

Theorem 8.11 (50.5, imbedding theorem)

Every compact metrizable space X of topological dimension m imbeds in $\mathbb{R}^{2m+1}$.

Proof. Let $N = 2m+1$, $|\cdot|$ the square metric on $\mathbb{R}^N$, and $\mathcal{C}(X, \mathbb{R}^N)$ with the sup metric ρ, complete and hence Baire. For continuous f let $\Delta(f) = \sup\{\operatorname{diam} f^{-1}(z) : z \in f(X)\}$ and $U_\epsilon = \{f : \Delta(f) < \epsilon\}$.

U_ϵ is open: let $\Delta(f) < b < \epsilon$. The set $A = \{(x, y) : d(x, y) \geq b\}$ is compact and $|f(x) - f(y)| > 0$ on it, with positive minimum δ (if $A \neq \emptyset$). If $\rho(f, g) < \delta/2$, then $g(x) \neq g(y)$ on A, so $\Delta(g) \leq b < \epsilon$.

U_ϵ is dense: let f and $\delta > 0$ be given. Cover X by open sets of diameter less than $\epsilon/2$ whose images under f have diameter less than $\delta/2$; by compactness and $\dim X = m$ there is a finite open refinement $\{U_1, \dots, U_n\}$ covering X of order at most $m+1$, and a partition of unity $\{\phi_i\}$ dominated by it (Theorem 4.16). Choose $x_i \in U_i$ and, by Lemma 8.10, points z_i in general position with $|z_i - f(x_i)| < \delta/2$. Let $g = \sum \phi_i z_i$. If $\phi_i(x) > 0$, then $x \in U_i$ and $|f(x) - z_i| < \delta$; since $f(x) - g(x) = \sum \phi_i(x)(f(x) - z_i)$, $|f(x) - g(x)| < \delta$, and by compactness $\rho(f, g) < \delta$. If $g(x) = g(y)$, then $\sum(\phi_i(x) - \phi_i(y)) z_i = 0$ with coefficients summing to 0, and only indices with $x \in U_i$ or $y \in U_i$ appear, at most $2(m+1) = N+1$ of them. By general position all coefficients vanish. Choosing i with $\phi_i(x) > 0$ gives $\phi_i(y) > 0$, so $x, y \in U_i$ and $d(x, y) < \epsilon/2$. Hence $\Delta(g) \leq \epsilon/2$ and $g \in U_\epsilon$.

By Lemma 8.1, $\bigcap_n U_{1/n}$ is dense, hence nonempty; a map g in it is injective, hence an imbedding (Theorem 3.20). ∎

Theorem 8.12 (50.6)

Every compact subspace of $\mathbb{R}^N$ has topological dimension at most N.

Proof.

Step 1. an open covering of $\mathbb{R}^N$ by sets of square-metric diameter at most 1, of order at most $N+1$. The cubical cells of $\mathbb{R}^N$ are the products $\sigma = \prod I_i$ with each I_i either $\{k_i\}$ ($k_i \in \mathbb{Z}$) or $(k_i, k_i + 1)$; let $P(\sigma)$ be the set of indices of the first kind. Every point lies in exactly one cell. Write $t_i(y)$ for the distance from y_i to $\mathbb{Z}$, and let

$$V(\sigma) = \Big\{y : \ |y_i - k_i| < \tfrac{1}{2} \ (i \in P(\sigma)), \quad y_i \in (k_i, k_i + 1) \ (i \notin P(\sigma)), \quad \max_{i \in P(\sigma)} t_i(y) < \min_{i \notin P(\sigma)} t_i(y)\Big\},$$

an open set of diameter at most 1 containing σ. If $\sigma \neq \tau$ and $|P(\sigma)| = |P(\tau)|$, then $V(\sigma) \cap V(\tau) = \emptyset$: if $P(\sigma) = P(\tau)$ some k_i differs, which is impossible on a point coordinate (nearest integer) and on

an interval coordinate (disjoint intervals); otherwise there are $i \in P(\sigma) - P(\tau)$ and $j \in P(\tau) - P(\sigma)$, and a common point would satisfy $t_i < t_j$ and $t_j < t_i$. So each point lies in at most one $V(\sigma)$ for each value $|P(\sigma)| \in \{0, \dots, N\}$, hence in at most $N+1$ of them.

Step 2. let $X \subset \mathbb{R}^N$ be compact and $\mathcal{A}$ an open covering of X, with Lebesgue number δ (Lemma 3.28). Choose $0 < s < \delta$; the sets $sV(\sigma) \cap X$ have diameter less than δ, so each lies in an element of $\mathcal{A}$; they form an open refinement covering X of order at most $N+1$. ■

Corollary 8.13 (50.7)

Every compact m-manifold has topological dimension at most m.

Proof. Each x has an open U with a homeomorphism $h : U \to W \subset \mathbb{R}^m$ open. Choose a closed ball $\overline{B} \subset W$ about $h(x)$; then $C_x = h^{-1}(\overline{B})$ is compact (so closed in X), contains the neighbourhood $h^{-1}(B)$ of x, and has dimension at most m by Theorem 8.12. Finitely many C_x cover X; apply Corollary 8.9. ■

Corollary 8.14 (50.8)

Every compact m-manifold imbeds in $\mathbb{R}^{2m+1}$.

Proof. It is compact Hausdorff, hence regular, with a countable basis, so metrizable (Theorem 4.12); its dimension k is at most m, and it imbeds in $\mathbb{R}^{2k+1} \subset \mathbb{R}^{2m+1}$ by Theorem 8.11. ■

Corollary 8.15 (50.9)

A compact metrizable space imbeds in some $\mathbb{R}^N$ iff it has finite topological dimension.

Proof. ($\Rightarrow$) Theorem 8.12.

($\Leftarrow$) Theorem 8.11. ■

Without compactness: every m-manifold, and more generally every metrizable space of dimension m with a countable basis, imbeds in $\mathbb{R}^{2m+1}$ (Menger–Nöbeling).